國 立 清 華 大 學

National Tsing Hua University

碩 士 論 文

Master's Thesis

基於自我確定性評估與熵門控的知識檢索

SEEK: Self-certainty Evaluated Entropy-gated Knowledge-retrieval

系所別： 資訊系統與應用研究所

Dept. / Grad. Inst.: Institute of Information Systems and Applications

學號 Student ID: S113065528

研究生 Author： 簡宏諭 Hung-Yu Chien

指導教授 Advisor： 高宏宇 博士 Dr. Hong-Yu Kao

中 華 民 國 一一五 年 七 月

July 2026

110022999

請替換成自己已簽名的授權書！！

國立清華大學學位論文授權書

Authorized Agreement for Thesis/Dissertation

(裝訂於紙本論文書名頁之次頁 For author to bind after the title page)

● 立書人（即論文作者）Author:　圖書館測試帳號　（以下稱本人 hereinafter referred to as " I "）

● 授權標的 Authorized subject：本人於　國立清華大學　（以下稱學校）理學院　物理研究所（研究所、學位學程）115學年度　第1學期之 □碩士 ■博士 學位論文
I hereby authorize that my □ Master’ s ■ Doctoral thesis submitted in the 1 semester of the 115 academic year at National Tsing Hua University (hereinafter referred to as "the University"), College of Science College, 物理研究所 Department / Graduate Institute / Degree Program,
論文題目 Title：清華大學圖書館博碩士論文庫測試
指導教授 Advisor：林登松

（以下稱**本著作**，本著作並包含論文全部、摘要、目錄、圖檔、影音以及相關書面報告、技術報告或專業實務報告等，以下同
Hereinafter referred to as "the publication", which contains all thesis/dissertation, abstracts, catalogues, graphic documents, audiovisual reports, technical reports or professional practice reports, etc.）

緣依據**學位授予法**等相關法令，對於本著作及其電子檔，學校圖書館得依法進行保存等利用，而國家圖書館則得依法進行保存、以紙本或讀取設備於館內提供公眾閱覽等利用。此外，為促進學術研究及傳播，本人在此並進一步同意授權(1)**國立清華大學**與(2)**國家圖書館對**本著作進行以下各點所定之利用：

In accordance with the Degree Conferral Act and other relevant laws and regulations, for this publication and its electronic file, the school library can be preserved and used according to the law. The National Central Library must preserve it in accordance with the law and permit public access in the library with paper or reading equipment. In addition, in order to promote academic research and scholarly communication, I hereby agree to authorize (1)**National Tsing Hua University** and the (2)**National Central Library** to use this publication for the following purposes

國立清華大學 NATIONAL TSING HUA UNIVERSITY

2026/04/02 09:16:10

請替換成自己的授權書!

110022999

一、授權部分：

本人**同意**授權國立清華大學與國家圖書館，無償、不限期間與次數重製本著作並得為教育、科學及研究等**非營利用途之利用**，其包括得將本著作之電子檔收錄於數位資料庫，並透過自有或委託代管之伺服器、網路系統或網際網路向位於全球之使用者(包含但不限於國立清華大學校園內、校園外及國家圖書館館內、館外)公開傳輸，以供使用者為非營利目的之檢索、閱覽、下載及/或列印。

I hereby agree to authorize National Tsing Hua University and the National Central Library to reproduce this publication free of charge, without limitation on time or the number of reproductions, and for education, science, research, and other non-profit use. This involves recording the electronic file of this publication into a digital database and the public transmission of them via self-owned or entrusted servers, network systems or the Internet to the user for the purpose of search, reading, downloading or printing for non-profit purposes.

二、本授權書第一點所定授權，均為非專屬且非獨家授權之約定，本人仍得自行或授權任何第三人利用本著作。

The authorized agreement is non-exclusive permission. I retain the right to use the publication myself or to authorize any third party to use it.

三、本授權書第一點所定授權對象，依各該點授權利用本著作時，均應尊重本人著作人格權及權利管理電子資訊等相關權利，不得以任何方式省略、增修或變更本人署名、本著作名稱、本著作內容及相關資料（包括本人原記載取得學位論文之學校全銜、書目等詮釋資料等）。

The authorized parties specified in this authorized agreement, when using this publication, shall respect my moral rights and the rights related to the management of electronic information. They shall not omit, alter, or modify my authorship, the title of the publication, the content of the publication, or related information (including the full name of the school where I obtained my degree, bibliographic details, and other interpretative materials) in any way.

四、依本授權書第一點將本著作及其電子檔對外公開之時間 Public date of the publication：

☐ 於完成審核後立即對外公開 Public immediately after completion of the review

☐ 於完成審核之六個月後對外公開 Public six months after completion of the review

■ 本人要求本著作應自 Public access after 民國 118 年 04 月 26 日起始得對外公開，故因本授權書第一點所定授權亦應自該日起始生效力。

國立清華大學 NATIONAL TSING HUA UNIVERSITY

請替換成自己已簽名的授權書!

110022999

五、本授權書第一點所定各該授權對象，均應各自遵守其授權範圍及相關約定。如有違反，由該違反之行為人自行承擔一切法律責任
The authorized party of this authorized agreement shall comply with the scope of authorization and relevant agreements. In case of any violation, the person responsible for the violation shall bear all legal responsibilities."

六、本人擔保本著作為本人創作而無侵害他人著作權或其他權利。如有違反，本人願意自行承擔一切法律責任
I guarantee that this publication was created by me without infringing copyright or other rights of others. In case of any violation, I am willing to assume full legal responsibility.

七、個資利用同意條款 Consent for the Use of Personal Data：
本人同意，學校及國家圖書館為本授權書所定各授權事項目的範圍內得蒐集、處理及利用本人所提供之個人資料，學校並可將該等個人資料提供給包括國家圖書館在內之相關第三人在同一目的範圍內處理及利用
I agree that National Tsing Hua University and the National Central Library may collect, process, and use the personal data I provide within the scope of the authorized matters specified in this authorized agreement. National Tsing Hua University may also provide such personal data to relevant third parties, including the National Central Library, for processing and use within the same purpose scope.

立授權書人 Signature of the author ：

(Date in ROC :Year/MM/DD)



國立清華大學
NATIONAL TSING HUA UNIVERSITY

2026/04/02 09:16:10

國立清華大學◯士學位論文
指導教授推薦書
National Tsing Hua University Thesis/Dissertation
Advisor Approval Form

◯◯學系(研究所)
◯◯◯ 君（學號◯◯◯◯）所提之論文

◯◯◯◯◯◯◯◯(中文論文題目)
經由本人指導撰述，同意提付審查。
Department of ◯◯◯◯
Mr. ◯◯ (Student ID: ◯◯◯◯) who has submitted the Thesis/Dissertation
◯◯◯◯◯◯◯◯(英文論文題目)
under my guidance, I approve for the submission to the oral defense committee.

□論文題目與內容符合本系(所、班、學位學程)專業領域
The subject and content of the thesis/dissertation are conformed to the professional field of the department (institute, class or program).

□符合本系(所、班、學位學程)論文相似度比對標準
The thesis/dissertation meets the standard for the thesis/dissertation originality check set by the department (institute, class or program).

指導教授 (簽章)
(Advisor) _____ (Signature)

中華民國 114 年 月 日
2025 / / (YYYY/MM/DD)

國立清華大學◯士學位論文
考試委員審定書
National Tsing Hua University
Final Thesis/Dissertation Review Form

◯◯學系(研究所)
◯◯◯ 君（學號◯◯◯◯）所提之論文

◯◯◯◯◯◯(中文論文題目)

經本委員會審查，符合◯士資格標準。

Department of ◯◯◯◯
Mr. ◯◯◯◯ (Student ID: ◯◯◯◯) who has submitted the Thesis/Dissertation

◯◯◯◯◯◯(英文論文題目)

has passed the oral defense and has met the qualifications to be awarded the degree of ◯◯◯◯.

學位考試委員會
(Oral defense Committee)

主持人 (簽章)
(Chair) _____(Signature)

委　員
(Members) _____

_____

_____

_____

中華民國 114 年 月 日
2025 / / (YYYY/MM/DD)

Abstract

This study proposes SEEK (Self-certainty Evaluated Entropy-gated Knowledge-retrieval), a lightweight adaptive Retrieval-Augmented Generation (RAG) framework for open-domain question answering. Standard RAG systems apply a fixed retrieval strategy to every query, which can inject noise on simple questions and miss intermediate evidence on multi-hop questions. Existing adaptive methods either route at a coarse question level, probe uncertainty at a noisy token level, or learn expensive sub-question policies through tree search and preference optimisation.

SEEK uses a compact KG-triple decomposer trained with a single LoRA SFT pass on a 12K mixture of MuSiQue, 2WikiMultihopQA, and PopQA examples. The final system no longer treats entropy as a single hard skip gate. Instead, it generates both a direct answer and a decomposed answer, preserves original-question evidence inside decomposed sub-question contexts, and applies a training-free verifier with a conservative type-aware fallback. This late-routing design preserves the interpretability of entropy-based methods while avoiding the brittleness of a global threshold.

Under the honest wiki-21M retrieval protocol aligned with DeepRAG, SEEK reaches an average of 40.52 EM and 47.61 F1, exceeding DeepRAG's 38.45 EM and 45.75 F1. SEEK improves over DeepRAG on 2WikiMultihopQA (50.00 EM), MuSiQue (24.70 EM), and PopQA (51.90 EM), while remaining behind on HotpotQA (35.50 EM versus 40.70 EM). Analysis shows that direct and decomposed paths are complementary, that oracle direct-vs-decompose routing still offers 3.8 to 6.1 EM of headroom, and that entropy is useful as a sub-question retrieval signal but not as a correctness guarantee.

Keywords: Retrieval-Augmented Generation, Adaptive Retrieval, Multi-hop Question Answering, Entropy, Verifier Routing, Large Language Models

摘要

本研究提出 SEEK (Self-certainty Evaluated Entropy-gated Knowledge-retrieval)，一個用於開放領域問答的輕量化自適應檢索增強生成 (Retrieval-Augmented Generation, RAG) 框架。傳統 RAG 對所有問題採用固定檢索策略：簡單問題可能因不必要的檢索引入雜訊，多跳問題則可能因單次檢索無法找到中介證據而回答失敗。既有自適應方法不是在問題層級做過於粗糙的路由，就是在詞元層級使用嘈雜的不確定性訊號，或像 DeepRAG 一樣透過樹搜尋與偏好最佳化學習昂貴的子問題決策策略。

SEEK 使用一個以 KG-triple 格式表示的分解器，透過單次 LoRA SFT 在 12K 筆 MuSiQue、2WikiMultihopQA 與 PopQA 混合資料上訓練。最終版本不再把熵值當成單一硬性 skip gate，而是同時產生 direct answer 與 decomposed answer，並將 original-question evidence 加回 decomposed sub-question context，最後由 training-free verifier 搭配保守的 type-aware fallback 選擇最終答案。這種 late-routing 設計保留了熵訊號的可解釋性，同時避免單一全域閾值在跨資料集時的不穩定。

在與 DeepRAG 對齊的 honest wiki-21M 檢索設定下，SEEK 平均達到 40.52 EM 與 47.61 F1，高於 DeepRAG 的 38.45 EM 與 45.75 F1。分資料集來看，SEEK 在 2WikiMultihopQA 達到 50.00 EM、MuSiQue 達到 24.70 EM、PopQA 達到 51.90 EM，皆高於 DeepRAG；HotpotQA 則爲 35.50 EM，仍低於 DeepRAG 的 40.70 EM。分析顯示 direct path 與 decomposition path 具有互補性，oracle direct-vs-decompose routing 仍有 3.8 到 6.1 EM 的提升空間，而 entropy 適合作爲 sub-question retrieval signal，但不能被視爲 correctness guarantee。

關鍵字：檢索增強生成、自適應檢索、多跳問答、熵、驗證器路由、大型語言模型

Acknowledgements

I would like to express my sincere gratitude to my advisor for the guidance and support throughout this research. I also thank the members of the Intelligent Knowledge Management Lab at National Tsing Hua University, the thesis oral examination committee, and all those who provided computational resources and helpful discussions. Finally, I thank my family and friends for their continuous encouragement.

Contents

學位論文授權書

指導教授推薦書

考試委員審定書

Abstract **i**

摘要 **ii**

Acknowledgements **iii**

Contents **vii**

List of Figures **viii**

List of Tables **x**

1 Introduction **1**

1.1 Background and Motivation 1

1.2 Research Objectives 1

2 Related Work **3**

2.1 Adaptive RAG 3

2.2 Reasoning and Knowledge Boundaries in RAG 8

2.3 Evaluation Datasets 11

3 Methodology **13**

3.1 Empirical Design Motivation 13

3.2 Overview 13

3.3 Problem Formulation 15

3.4 Stage 1: Sub-Question Decomposer . . . 16
3.4.1 Output Format . . . 16
3.4.2 Decomposer Prompt . . . 16
3.4.3 Decomposition Examples by Dataset . . . 17
3.5 Stage 2: Token-Level Confidence Probe . . . 17
3.6 Stage 3: Decomposition-Path Retrieval Gate . . . 18
3.6.1 Original-Question Evidence Union . . . 18
3.6.2 Retriever . . . 19
3.7 Stage 4: Synthesis . . . 19
3.8 Stage 5: Training-Free Verifier v2 . . . 20
3.9 Training Protocol . . . 21
3.9.1 Backbone and LoRA Configuration . . . 21
3.9.2 Training Data . . . 21
3.9.3 Optimisation . . . 22
3.10 Inference Modes . . . 22
3.11 Implementation Summary . . . 22

4 Experiments **24**
4.1 Experimental Setup . . . 24
4.1.1 Datasets . . . 24
4.1.2 Retrieval Corpus . . . 24
4.1.3 Metrics and Baselines . . . 25
4.1.4 Implementation Details . . . 25
4.2 Main Results . . . 25
4.3 Retrieval Recall . . . 27
4.4 Gold and Base Decomposition on MuSiQue . . . 27
4.5 Retrieval Is Necessary After Decomposition . . . 29
4.6 Direct-Only Confidence-Gated Baseline . . . 30
4.7 Threshold Sensitivity . . . 31
4.8 Path-Aware Title Retrieval . . . 32
4.9 Original-Question Evidence Union . . . 32

4.10 Final Path Selection Rate . . . 33
4.11 Direct-vs-Decompose Correctness Distribution . . . 34
4.12 Decomposition-Path Failure Taxonomy . . . 34
4.13 Case-Level Comparison with DeepRAG . . . 36
4.14 Oracle Direct-vs-Decompose Headroom . . . 38
4.15 Entropy as a Confidence Signal . . . 38
4.16 Distractor-Pool Development Results . . . 42
4.17 Summary . . . 43

5 Discussion **44**
5.1 Cost–Accuracy Trade-off versus DeepRAG . . . 44
5.2 Why Direct and Decomposition Paths Must Coexist . . . 45
5.3 What Training the Decomposer Buys . . . 46
5.4 Original-Question Evidence Union . . . 47
5.5 Entropy: Useful Signal, Not Correctness Guarantee . . . 47
5.6 Remaining Headroom and Remaining Problems . . . 48
5.7 Threats to Validity . . . 49
5.7.1 External Baseline Numbers Are Paper Numbers . . . 49
5.7.2 Dataset Slicing . . . 49
5.7.3 Single Seed and Greedy Decoding . . . 50
5.7.4 Routing Cannot Fix Both-Wrong Cases . . . 50

6 Conclusions and Future Work **51**
6.1 Summary of Contributions . . . 51
6.2 Limitations . . . 52
6.3 Future Work . . . 53
6.3.1 Stronger Verifier Signals . . . 53
6.3.2 Better Retrieval and Reranking . . . 53
6.3.3 Length-Conditioned Decomposer . . . 53
6.3.4 Multi-Sample Decomposition . . . 53
6.3.5 Evidence-Aware Verification . . . 54
6.3.6 Span Calibration . . . 54

6.4 Closing Remarks 54

References **55**

Appendices

A Supplementary Material **60**

List of Figures

3.1 SEEK Pipeline 15

4.1 Answer-Containing Retrieval Recall Figure 28
4.2 Three-Way Retrieval Ablation 30
4.3 Direct-vs-Decompose Correctness Counts 35
4.4 Oracle Routing Headroom 39
4.5 Confidence by Correctness Quadrant 40

List of Tables

2.1 Comparison of Adaptive RAG Methods 8

3.1 Empirical Motivation for SEEK Design 14
3.2 Decomposition Patterns by Dataset 17
3.3 Decomposer Training Data Composition 21

4.1 Main Results on Honest Wiki-21M 26
4.2 Answer-Containing Retrieval Recall 27
4.3 MuSiQue Gold and Base Decomposition Recall 29
4.4 Retrieval Necessity After Decomposition 29
4.5 Direct-Only Confidence-Gated Baseline 31
4.6 Threshold Sensitivity 31
4.7 Title-Field Matching Ablation 32
4.8 Original-Question Evidence Union Ablation 33
4.9 Final Direct-vs-Decompose Selection Rate 33
4.10 Correct Answer Distribution Across Two Paths 34
4.11 Decomposition-Path Failure Taxonomy 35
4.12 DeepRAG-vs-SEEK Correctness Quadrants 36
4.13 DeepRAG Failure Signals in the SEEK-Win Set 37
4.14 Oracle Direct-vs-Decompose Headroom 38
4.15 Confidence by Direct-vs-Decompose Quadrant 39
4.16 Direct-Path Accuracy by Entropy Quintile 40
4.17 Confidence Signal Battery 42
4.18 Distractor-Pool Development Results 43

5.1 Inference Retrieval Count Comparison 44

5.2 Offline Adaptation Cost Comparison 45

Chapter 1 Introduction

1.1 Background and Motivation

Large language models (LLMs) often hallucinate when queried about recent, obscure, or proprietary information outside their training scope. Retrieval-Augmented Generation (RAG) addresses this by fetching relevant external evidence, reducing factually unsupported answers [1], [2]. However, applying RAG indiscriminately creates two problems. For simple questions, unnecessary retrieval can inject distracting evidence and increase latency. For complex multi-hop questions, a single retrieval step often fails to surface the intermediate entities needed for reasoning.

Adaptive RAG therefore needs to decide not only *whether* retrieval is useful, but also *when to trust decomposition*. Decomposition can sharpen retrieval queries, but it can also remove the original question's comparison targets, entity pairs, or disambiguating clues. This thesis frames adaptive RAG as a *decomposition-routing problem*: the system should control retrieval at the sub-question level while preserving the original-question path as a competing candidate.

1.2 Research Objectives

This thesis introduces SEEK (Self-certainty Evaluated Entropy-gated Knowledge-retrieval), a lightweight adaptive RAG framework built around training-free verifier routing. The objectives are:

- Build a compact KG-triple decomposer that converts multi-hop questions into atomic `entity >> relation` sub-questions.
- Use token-level entropy as an interpretable confidence signal, while showing that entropy alone is insufficient as a global hard router.

- Generate both direct and decomposed candidate answers, preserve original-question evidence inside decomposed steps, and select the final answer with a training-free verifier based on confidence, answer type, decomposition stability, and candidate agreement.
- Evaluate the final system under the honest wiki-21M retrieval protocol used by DeepRAG, showing where a lightweight verifier can match or exceed a learned retrieval policy and where routing headroom remains.

Chapter 2 Related Work

2.1 Adaptive RAG

Retrieval-Augmented Generation (RAG) [1] augments a parametric language model with a non-parametric retriever, allowing the model to condition its generation on external evidence rather than relying only on parameters learned during pretraining. A typical RAG pipeline can be described as five stages:

1. **Indexing.** The source corpus is segmented into passages, and each passage is stored in a form that supports retrieval, such as sparse lexical indexes or dense vector representations.

2. **Query manipulation.** The input question may be rewritten, expanded, normalised, or decomposed so that the retriever can better match it with relevant evidence.

3. **Retrieval.** The system selects candidate passages from the indexed corpus, commonly using sparse lexical matching, dense vector similarity, or a hybrid retrieval strategy.

4. **Post-retrieval processing.** The initially retrieved passages may be filtered, reranked, compressed, or otherwise refined before they are passed to the generator.

5. **Generation.** The language model conditions on the selected evidence and produces the final answer, ideally grounding its output in the retrieved passages.

This retrieve-then-generate paradigm is now widely used to make language model outputs more factual in question answering, explanation writing, and other knowledge-intensive tasks. Its central advantage is that factual evidence can be updated externally without retraining the language model. For multi-hop question answering, however, the quality of the retrieved evidence depends strongly on the query that is issued at each step: a single original question may not contain the intermediate entity needed to retrieve the

next supporting passage.

This fixed retrieval assumption creates two well-documented failure modes:

- **Noise injection on easy queries.** For questions the model already knows (e.g. "Who wrote *Hamlet*?"), retrieved passages introduce distractor entities that can override correct parametric knowledge [2].
- **Insufficient evidence on hard queries.** For multi-hop questions, a single retrieval pass rarely surfaces all bridging entities, leading to incomplete or compositionally wrong answers [3].

These two failure modes motivate adaptive retrieval: deciding **when** retrieval is useful, **how much** evidence to fetch, and what signal should determine the gate.

Adaptive RAG methods decide whether and how much to retrieve for each query or reasoning step. They can be characterised along three axes: decision granularity, gating signal, and training cost. Decision granularity ranges from question-level routing to sentence-, token-, or sub-question-level decisions, while training cost ranges from training-free methods to supervised fine-tuning, reinforcement learning, or preference optimisation. Since the gating signal is the most consequential axis for SEEK, we organise existing methods into three families: classifier-based gating, confidence-based gating, and reasoning-step decision methods.

Classifier-based gating. The first family trains a dedicated discriminator that inspects the query and emits a retrieval decision.

Adaptive-RAG [4] routes each incoming query to one of three retrieval strategies according to its predicted complexity. Easy questions are answered without retrieval, medium-complexity questions use a single retrieval step, and complex questions are routed to a multi-step retrieve-and-reason pipeline such as **IRCoT** [3]. The router is a T5-Large classifier trained with automatically derived labels. Each training question is assigned to the cheapest strategy that still produces the correct answer, so the classifier learns a cost-aware mapping from question complexity to retrieval depth. This design reduces

unnecessary retrieval compared with always using a multi-step pipeline. However, the decision is made only once at the beginning of the query. As a result, the system cannot react to uncertainty that appears during intermediate reasoning, and its routing labels remain coarse: questions with different numbers of reasoning hops may still be assigned to the same multi-step category.

SKR [5] approaches adaptive retrieval from the perspective of self-knowledge. Instead of predicting general question complexity, SKR estimates whether the language model is likely to know the answer from its own parametric memory. If the question is judged as known, the model answers directly; if it is judged as unknown, external retrieval is invoked. This judgement can be implemented with a trained classifier or with memory-based self-knowledge strategies. In the classifier setting, SKR is a binary retrieval gate: it decides whether to retrieve, but does not distinguish between single-step and multi-step retrieval. Therefore, compared with Adaptive-RAG, SKR gives a more direct estimate of whether retrieval is needed, but it still commits to a question-level decision before reasoning begins.

Confidence- and uncertainty-based gating. The second family replaces an external router with uncertainty signals observed during generation. These methods are often training-free, since the retrieval decision is derived from token probabilities, entropy, attention patterns, or internal model states rather than from an additional trained classifier.

FLARE [6] moves the retrieval decision from the question level to the generation level. At each step, it first drafts a tentative next sentence and checks whether the sentence contains low-confidence tokens. If the confidence falls below a threshold, the tentative sentence is used to formulate a retrieval query, and the model regenerates the sentence with the retrieved evidence. This design allows retrieval to occur only when the model appears uncertain during generation. However, token probability is only an indirect proxy for factual uncertainty. A low-probability token may reflect lexical variation rather than missing knowledge, and sentence-level regeneration may not align well with multi-hop reasoning, where the natural decision unit is often a sub-question.

DRAGIN [7] further refines dynamic retrieval by considering both when to retrieve and what to retrieve. Its *Real-time Information Need Detection*(RIND) module estimates whether the current generation state requires external evidence, while its *Query Formulation based on Self-attention*(QFS) module constructs retrieval queries from salient context tokens instead of relying only on the most recent sentence. Compared with FLARE, DRAGIN provides a more context-aware retrieval trigger and query formulation strategy. Nevertheless, its decision is still made at the token-generation level, which can be mismatched with explicit sub-question reasoning, and it introduces additional overhead from attention-based query construction.

SeaKR [8] estimates retrieval need from the model's internal states rather than from output probabilities alone. It activates retrieval when the model exhibits high self-aware uncertainty and reranks retrieved passages according to how much they reduce that uncertainty. This makes the retrieval signal more closely tied to the model's internal generation dynamics than token-level probability alone. However, the signal is still an estimate of uncertainty rather than a direct measure of answer correctness. A model may be uncertain for reasons unrelated to missing external knowledge, and retrieval may not necessarily improve the answer even when uncertainty is high.

TARG [9] studies adaptive retrieval as a training-free, single-shot decision. It generates a short no-context draft and computes uncertainty from the prefix logits to decide whether retrieval should be invoked. In addition to entropy, TARG considers a top-1/top-2 logit-margin signal and a small-sample variance signal. Its main strength is that it provides a lightweight retrieval gate without training an auxiliary classifier or modifying the language model. However, its evaluation mainly focuses on open-domain factoid QA settings, where each question is typically answered with a single retrieval decision. Its behaviour in multi-hop QA or decomposition-based retrieval settings therefore remains less explored.

Across this family, uncertainty provides a useful and interpretable signal for adaptive retrieval, but it remains an imperfect proxy. Low uncertainty does not guarantee a correct answer, and high uncertainty does not always indicate that retrieval will help. SEEK addresses this limitation by using entropy as a sub-question retrieval signal and as one

component of a late verifier, rather than treating it as a single hard router for the entire question.

Learned retrieval-policy methods. The third family does not rely on a fixed classifier or a hand-set uncertainty threshold. Instead, it learns retrieval decisions as part of the model's reasoning or generation process.

Self-RAG [10] trains the language model to generate special reflection tokens that control retrieval and evaluate retrieved passages. For example, the model can emit `[Retrieve]` to decide whether external evidence is needed, and then use additional tokens to judge whether a passage is relevant, supportive, or useful. This design makes retrieval part of generation itself, rather than a separate preprocessing step. However, it requires fine-tuning on reflection-token supervision and a custom decoding procedure that interleaves normal text generation with retrieval-related decisions.

DeepRAG [11] formulates retrieval-augmented reasoning as a step-by-step decision process. At each sub-query, the model chooses between answering from parametric memory and issuing a retrieval call. Its training procedure first uses binary tree search to explore both choices for each sub-query, and then applies preference-based calibration to learn which reasoning traces should be preferred. This gives DeepRAG strong sub-question-level retrieval control, but the cost is substantial. Tree-search rollout becomes expensive as the number of reasoning steps grows, preference pairs must be carefully constructed, and the final retrieval policy is learned implicitly rather than exposed as an interpretable score.

A related line of work studies retrieval cost more directly. **Fast-or-Better** [12] treats the number of retrieved passages or retrieval calls as a controllable trade-off between latency and accuracy, rather than assuming one fixed operating point. This view reinforces an important point for adaptive RAG: retrieval should be evaluated not only by accuracy, but also by the cost required to obtain that accuracy.

SEEK is closest in spirit to DeepRAG because both methods make retrieval decisions at the sub-question level. The key difference is that SEEK avoids tree search and

preference optimisation. It uses a single supervised decomposer and replaces the learned retrieval policy with a training-free verifier based on entropy, decomposition stability, answer type, and candidate agreement.

Table 2.1 summarises the key differences between SEEK and existing adaptive RAG methods along the three axes introduced above. SEEK occupies the under-explored region of *late verifier-based routing with interpretable, training-free signals*: only a single SFT pass is needed for the decomposer, and the final route is decided by transparent query-level diagnostics rather than a learned policy.

Table 2.1: Comparison of adaptive RAG methods along granularity, gating signal, interpretability, training cost, and decision unit.

Method	Decomposer	Gating Signal	Interpretable	Training Cost	G
Adaptive-RAG [4]	None	Query classifier	Partial	Low (T5-Large)	
FLARE [6]	None	Token probability	Yes	None	
Self-RAG [10]	None	Reflection tokens	Partial	High (distill)	
DRAGIN [7]	None	Entropy + attention	Yes	None	
SKR [5]	None	Binary self-knowledge	Partial	Low	
SeaKR [8]	None	Hidden-state variance	No	None	
TARG [9]	None	Prefix uncertainty	Yes	None	
DeepRAG [11]	SFT	Learned policy (SFT+DPO)	No	High	
SEEK (Ours)	SFT	Entropy-aware verifier	**Yes**	**Low** (SFT only)	**Sub**

2.2 Reasoning and Knowledge Boundaries in RAG

A parallel thread focuses on *how* to retrieve and reason once retrieval is deemed necessary, independent of the gate.

The foundational technique is **IRCoT** [3], which interleaves chain-of-thought reasoning with retrieval: after each generated CoT sentence, the partial reasoning becomes the next query. IRCoT is the canonical "multi-step" implementation referenced by **Adaptive-**

RAG [4] and **DeepRAG** [11], but its retrieval count grows linearly in the CoT length, regardless of whether intermediate steps are factually difficult. Question decomposition provides a complementary way to structure multi-hop QA. Rather than retrieving after each free-form reasoning sentence, decomposition-based methods explicitly split a complex question into simpler sub-questions that can be answered independently or sequentially. Early systems such as **DecompRC** [13] show that compositional questions can be reduced to single-hop reading-comprehension steps, while **BREAK/QDMR** [14] formalises decomposition as an ordered sequence of natural-language operations required to answer the original question. SEEK inherits this decomposition paradigm, but uses it primarily as an inspectable reasoning path: it preserves the original-question context for decomposed steps, evaluates whether the generated sub-questions are useful, and verifies whether the direct or decomposed answer should be trusted.

More recent work synthesises reasoning data to internalise the retrieve-and-reason loop. **Auto-RAG** [15] autonomously synthesises multi-turn retrieval–reasoning trajectories and fine-tunes on them, so that the model plans its own retrieval steps without hand-crafted prompts. **Search-o1** [16] embeds retrieval inside a large reasoning model in an agentic loop, augmenting o1-style long-chain reasoning with on-demand search and a document-reasoning module that distils retrieved passages before they re-enter the chain. **AirRAG** [17] activates the model's intrinsic reasoning by combining Monte Carlo Tree Search with self-consistency to explore alternative solution paths. These methods substantially strengthen the reasoning side of RAG, but — like DeepRAG — they rely on heavy data synthesis or search-time expansion; SEEK instead keeps the reasoning component a single lightweight SFT decomposer and concentrates its novelty in training-free verification.

Every adaptive method above ultimately rests on a model's ability to recognise its own *knowledge boundary* — the line between what it reliably knows and what it must look up. A dedicated line of work studies this capacity directly, and we group it into three strands: *self-knowledge elicitation*, *uncertainty estimation*, and the *interaction between the knowledge boundary and retrieval*.

Self-knowledge and its calibration. A first strand asks whether a model can introspect on what it knows. Kadavath et al. [18] show that LLMs are, to a useful degree, calibrated: a model can be prompted to predict the probability `P(True)` that its own answer is correct, and these self-judgements correlate with actual accuracy — though calibration degrades on out-of-distribution and adversarial questions. Yin et al. [19] construct a "SelfAware" benchmark of answerable and unanswerable questions and find that LLMs only partially "know what they don't know": self-knowledge can be elicited but lags well behind human performance. From the supervised side, Kapoor et al. [20] argue that models "must be taught to know what they don't know", demonstrating that lightweight calibration-tuning or probing substantially sharpens a model's estimate of its own uncertainty. Yin et al. [21] formalise the notion of a knowledge boundary itself, proposing to benchmark a model by probing the frontier of its reliable knowledge rather than by aggregate accuracy alone.

Uncertainty estimation as a boundary signal. A second strand asks how to *measure* the boundary at inference time. Beyond retrieval-specific methods, recent work on LLM uncertainty quantification compares several confidence estimators for answer correctness, including token entropy, maximum softmax probability, margin-based scores, verbalised confidence, and response consistency [22]. These studies suggest that no single confidence signal is universally reliable across tasks and models. This motivates evaluating whether entropy is sufficient in SEEK's QA setting, or whether a margin-based signal provides a more discriminative retrieval gate.

Semantic entropy further shows why token-level uncertainty can be limited for natural-language answers: different surface forms may express the same meaning, so uncertainty over token sequences does not always correspond to uncertainty over answers. Kuhn et al. [23] introduce *semantic entropy*, which clusters multiple sampled answers by meaning before computing entropy, so that paraphrases of one correct answer do not register as uncertainty; Farquhar et al. [24] scale this idea and show in *Nature* that semantic entropy reliably detects confabulations across QA and biography generation. This observation is important for SEEK because exact QA answers often involve aliases, entity variants, and short spans. We therefore treat token entropy as a cheap diagnostic signal rather than as a

correctness guarantee.

The boundary under retrieval. A third strand studies the boundary specifically in the RAG setting. Ren et al. [25] show that retrieval augmentation not only injects evidence but also *reshapes* the boundary: relevant passages let a model answer questions it could not before, while irrelevant or distracting passages can push a previously-correct answer over the boundary into error — precisely the "noise injection on easy queries" failure mode of §2.1. This makes a per-step boundary estimate, rather than a one-shot query-level one, essential when retrieval and reasoning are interleaved.

These three strands form the conceptual foundation of SEEK. The failure modes of §2.1 — spurious retrieval on easy queries, missing retrieval on hard ones, and confident hallucination under a hard entropy threshold — are all symptoms of a poorly localised knowledge boundary. Where confidence-based methods read a single uncertainty proxy and policy-based methods bury the boundary inside a learned policy, SEEK makes the boundary inspectable through a late verifier: entropy, decomposition stability, and answer agreement jointly decide whether the direct or decomposed path should be trusted.

2.3 Evaluation Datasets

The evaluation of adaptive RAG hinges on datasets that span both single-hop and multi-hop complexity. This thesis uses HotpotQA, 2WikiMultihopQA, MuSiQue, and PopQA to cover both multi-hop reasoning and single-hop factual recall.

HotpotQA. HotpotQA [26] is the canonical multi-hop QA benchmark, containing 113K Wikipedia-based questions requiring 2-hop reasoning. Each example provides (a) the question, (b) the gold answer, (c) two gold-supporting paragraphs out of 10 distractors (in the `distractor` setting), and (d) sentence-level supporting facts. Questions fall into two compositional patterns: *bridge* (the answer to sub-question 1 is the subject of sub-question 2) and *comparison* (two entities are compared along a shared attribute). HotpotQA is a primary training source for DeepRAG and a standard evaluation set for nearly all adaptive RAG papers.

2WikiMultihopQA. 2WikiMultihopQA [27] constructs 192K multi-hop questions by leveraging Wikidata knowledge-graph paths, ensuring each question requires traversal of an explicit reasoning chain. It includes four question types (comparison, inference, compositional, and bridge-comparison) and provides KG-triple annotations alongside textual evidence. Compared to HotpotQA, 2WikiMultihopQA exhibits less surface-form leakage between the question and the supporting passages, making it a more reliable test of compositional reasoning.

MuSiQue. MuSiQue (**Mu**ltihop questions via **Si**ngle-hop **Que**stions) [28] addresses two pathologies of earlier datasets: (i) shortcut answers via single-hop reasoning, and (ii) weak distractor selection. MuSiQue is built by composing 2–4 single-hop questions into chains using strict connectivity criteria, yielding 25K questions of 2-hop, 3-hop, and 4-hop varieties. Importantly, MuSiQue provides the *decomposed* sub-questions as supervision, which makes it a natural training source for decomposer models such as the one used in SEEK. Its high distractor difficulty makes it the hardest of the four datasets in this thesis.

PopQA. PopQA [29] is a *single-hop* entity-centric QA dataset of 14K questions, each derived from a Wikidata subject–property–object triple. Critically, PopQA tags each entity with its *popularity* (measured by monthly Wikipedia page views), enabling stratified evaluation of factual recall on long-tail entities. Despite being single-hop, PopQA exposes the over-decomposition failure mode of decomposer-based pipelines: trained on multi-hop data, a decomposer may emit two-step plans even when one step suffices, harming both EM and efficiency.

Single-hop open-domain QA. For completeness, prior adaptive-RAG papers also evaluate on **Natural Questions (NQ)** [30], **TriviaQA** [31], and **SQuAD** [32]. These are single-hop datasets whose answers typically exist in a single Wikipedia paragraph, and they primarily test whether an adaptive method can correctly *skip* retrieval rather than orchestrate it. This thesis follows the DeepRAG protocol of evaluating on PopQA as the representative single-hop test set.

Chapter 3 Methodology

This chapter presents **SEEK** (**S**elf-certainty **E**valuated **E**ntropy-gated **K**nowledge-retrieval), a lightweight adaptive RAG framework that treats question decomposition as a routing problem. Earlier versions of SEEK used entropy as a hard skip-decomposition gate. The final system instead runs both a direct path and a decomposition path, preserves original-question evidence inside decomposed steps, and chooses between the candidate answers using a training-free verifier. Section 3.1 summarises the empirical observations that motivated this design. Section 3.2 gives the end-to-end overview. Sections 3.4–3.7 describe the decomposition path, Section 3.8 describes the verifier, and Sections 3.9–3.10 give the training and implementation details.

3.1 Empirical Design Motivation

SEEK was developed from a set of empirical observations about where adaptive RAG succeeds and fails. These observations are evaluated in detail in Chapter 4; here they are used to motivate the final system design.

These observations imply that adaptive RAG is more than a retrieve-or-skip problem. A useful system must decide whether a question should be answered directly, decomposed into sharper retrieval queries, or repaired with original-question context after decomposition. This motivates the dual-path architecture used throughout the rest of the thesis.

3.2 Overview

Given a question q and an open-domain retrieval corpus $\mathcal{P}$, SEEK produces a final answer $\hat{a}$ through two candidate-generation paths followed by a verifier:

1. **Direct path.** The answerer directly answers the original question q with BM25 evidence retrieved from $\mathcal{P}$, producing a direct candidate a^{dir} and a token-level entropy

Table 3.1: Empirical observations that motivate the final SEEK design.

Design choice	Observation	Resulting decision
Use decomposition for retrieval	Decomposed queries improve answer-containing hit@5 by +10.0 on HotpotQA, +30.6 on 2Wiki, and +20.5 on MuSiQue.	Make sub-question retrieval the main multi-hop evidence acquisition path.
Retrieve after decomposition	Removing retrieval after decomposition loses 9.3 to 27.3 EM across datasets.	Treat decomposition as a query-planning step, not as a substitute for external evidence.
Avoid a single global entropy router	Maximum entropy is useful, but confident wrong answers remain: 9.5% to 28.8% of both-wrong questions look as confident as both-correct questions.	Use entropy as one verifier signal rather than as a hard whole-question skip gate.
Preserve original-question evidence	Short sub-questions can drop comparison targets or disambiguating clues; original-question evidence union improves the decomposed candidate by up to +5.5 EM.	Add original-question evidence to each decomposed step while keeping direct and decomposed answers separate.
Keep both answer paths	Direct-only and decompose-only wins coexist, especially on HotpotQA where the two columns are nearly tied.	Generate both candidates and perform late training-free verification.

score σ^{dir} over the generated answer span.

2. **Decomposition path.** The decomposer $\mathcal{D}_\theta$ maps q to ordered sub-questions $\{s_1, \ldots, s_n\}$. Each sub-question is answered sequentially, with placeholder substitution, entropy probing, conditional retrieval, original-question evidence union, and a final synthesis step that produces a^{dec}.

3. **Training-free verifier.** A deterministic verifier compares a^{dir} and a^{dec} using query-level signals: direct-answer confidence, decomposition stability, post-synthesis confidence, and answer agreement. It returns either a^{dir} or a^{dec} as $\hat{a}$.

The final operating point is the **training-free verifier with type-aware fallback**: it

uses only signals naturally emitted during inference and does not require an additional classifier, DPO, tree search, or gold-answer supervision. Evidence-support variants are evaluated as ablations in Chapter 4, but are not used in the main method because lexical support is not globally stable for compositional reasoning.

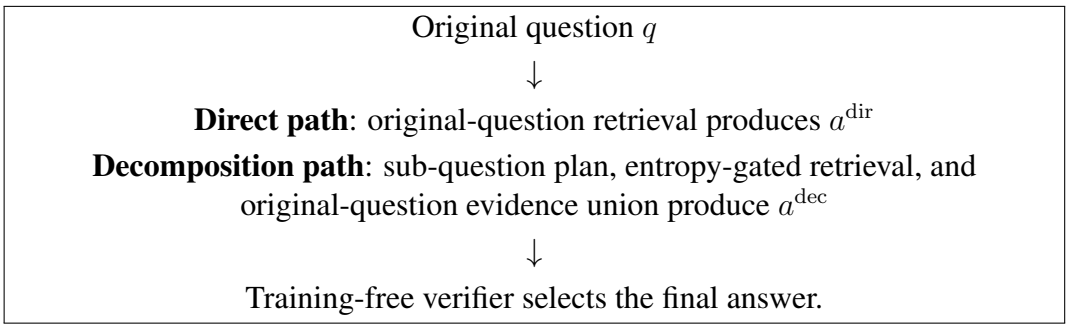


Fig. 3.1: SEEK pipeline: the system generates a direct answer and a decomposed answer, preserves original-question evidence inside the decomposition path, and uses a training-free verifier to select the final answer.

3.3 Problem Formulation

Let $\mathcal{M}_\theta$ be a pretrained language model, $r(q, \mathcal{P}, k)$ a BM25 retriever over an open-domain corpus $\mathcal{P}$, and $\mathcal{D}_\theta(q)$ a decomposer that emits sub-questions. A conventional retrieve-then-read system produces one answer from one path. SEEK instead constructs two candidates:

$$a^{\mathrm{dir}} = \mathcal{M}_\theta(q, r(q, \mathcal{P}, k)),$$

$$a^{\mathrm{dec}} = \mathrm{Synth}\left(q, \{(s_i, a_i)\}_{i=1}^{n}\right), \quad \{s_i\}_{i=1}^{n} = \mathcal{D}_\theta(q).$$

The final decision is a verifier v:

$$\hat{a} = v\left(q, a^{\mathrm{dir}}, a^{\mathrm{dec}}, z^{\mathrm{dir}}, z^{\mathrm{dec}}\right),$$

where z^{dir} and z^{dec} are inference-time diagnostic signals such as entropy, unresolved

placeholders, and candidate agreement. Unlike DeepRAG [11], SEEK does not learn v through tree-search rollouts or preference optimisation. The verifier is a transparent rule set designed to approximate the oracle choice between direct and decomposed reasoning.

3.4 Stage 1: Sub-Question Decomposer

3.4.1 Output Format

The decomposer emits one sub-question per line, prefixed with `#`$\langle i \rangle$`:`. Each sub-question is expressed as a compact knowledge-graph fragment:

$$\texttt{\#}i\texttt{:} \quad \langle \textit{entity} \rangle \;\; \texttt{>>} \;\; \langle \textit{relation} \rangle$$

When a sub-question depends on an earlier answer, the placeholder `#`j refers to step j. For example:

```
#1: X >> director
#2: #1 >> place of birth
```

At inference, placeholders are resolved sequentially: once a_1 is produced, `#1` in s_2 is rewritten as a_1 before retrieval and answering. The KG-triple notation is compact, consistent across datasets, and aligned with Wikidata-style supervision in 2WikiMultihopQA and PopQA.

3.4.2 Decomposer Prompt

The decomposer is prompted with a fixed instruction:

> *Decompose the following question into an ordered list of atomic sub-questions. Each sub-question must be answerable in a single retrieval and reasoning step. When a sub-question depends on the answer of an earlier one, refer to that step with a `#N` placeholder. Output one sub-question per line, prefixed with `#<index>:`*

Table 3.2: Representative decomposition patterns produced by the SEEK decomposer.

Dataset	Question type	Decomposition
PopQA	1-hop fact lookup	`#1: George Rankin >> occupation`
HotpotQA	2-hop bridge	`#1: Film X >> director` `#2: #1 >> nationality`
2WikiMultihopQA	comparison	`#1: A >> date of birth` `#2: B >> date of birth`

MuSiQue	compositional chain	`#1: X >> capital`
		`#2: #1 >> mayor`
		`#3: #2 >> party`

3.4.3 Decomposition Examples by Dataset

3.5 Stage 2: Token-Level Confidence Probe

Both the direct path and each sub-question answer use an answer-only prompt:

> *Answer the question with the answer phrase only. No explanation, no preamble, no full sentence — just the answer.*

During greedy decoding, SEEK records the softmax distribution $p_t = \mathrm{softmax}(\mathrm{logits}_t)$ for each generated token t and computes:

$$\sigma^{\text{max-ent}} = \max_t H(p_t), \tag{3.1}$$

$$\sigma^{\text{mean-ent}} = \frac{1}{T}\sum_t H(p_t), \tag{3.2}$$

$$\sigma^{\text{min-lp}} = \min_t \log p_t(y_t), \tag{3.3}$$

where $H(p) = -\sum_v p(v)\log p(v)$ and EOS tokens are excluded. The default diag-

nostic signal is maximum token entropy. In the decomposition path, this entropy controls per-step retrieval. In the v2 verifier, it is used at the query level to decide whether the direct answer is confident enough to rescue a failed decomposition.

3.6 Stage 3: Decomposition-Path Retrieval Gate

Within the decomposition path, retrieval remains sub-question-level. For a resolved sub-question s_i, the answerer first produces a tentative parametric answer $\tilde{a}_i$ and entropy σ_i. The retriever is invoked when:

$$\pi(s_i) = \mathbb{1}[\sigma_i \geq \tau],$$

with $\tau = 0.3$ for max-entropy in the final system. If $\pi(s_i) = 1$, BM25 retrieves top-k passages and the answerer re-answers s_i with context; otherwise $\tilde{a}_i$ is kept. This local gate reduces unnecessary retrieval inside the decomposition path, but it is no longer the final routing mechanism for the whole question.

3.6.1 Original-Question Evidence Union

A decomposed sub-question is often shorter and more entity-centric than the original question, which improves retrieval precision but can remove global clues such as comparison targets or disambiguating conditions. To preserve this context, the final system augments each retrieved sub-question context with evidence retrieved from the original question:

$$C_i = r(s_i, \mathcal{P}, k) \cup r(q, \mathcal{P}, k).$$

The union is deduplicated before answering. This operation does not change the decomposer or train a new retriever; it simply keeps the original-question evidence available when answering each decomposed step. Although these original-question passages are also the evidence source for the direct candidate, the two answer paths remain dis-

tinct: the direct path answers the original question directly, whereas the decomposition path still performs sub-question retrieval, step answering, and final synthesis. Thus the union shares evidence, not the direct answer or the final routing decision. Chapter 4 evaluates this operation as a decomposition-path ablation, especially on HotpotQA and 2WikiMultihopQA.

3.6.2 Retriever

The final experiments use an honest open-domain Wikipedia corpus: the shared Elasticsearch `wiki` index over `psgs_w100` with approximately 21M passages, matching DeepRAG's retrieval setting. We use BM25 top-$k = 8$ for HotpotQA, 2WikiMultihopQA, and MuSiQue, and top-$k = 3$ for PopQA. Earlier per-question distractor-pool results are retained only as development and ablation evidence. The final implementation uses a path-aware title policy: title matching is disabled for long original-question direct retrieval and enabled for short decomposed sub-question retrieval.

3.7 Stage 4: Synthesis

The decomposed answer a^{dec} is produced by a synthesis pass over the original question and all (s_i, a_i) pairs. The prompt encodes three answer-shaping rules derived from error analysis:

1. **Comparison questions** return the matching entity, not the compared value.
2. **yes-or-no questions** return "Yes" or "No" based on agreement between sub-answers.
3. **Terse-span rule** strips unnecessary modifiers and keeps answer spans short for EM compatibility.

The implementation also includes a value-to-entity guard for comparison failures: if synthesis emits a value such as a year but the question asks which entity satisfies a comparison, the system maps the value back to the owning entity when this mapping is unambiguous.

3.8 Stage 5: Training-Free Verifier v2

The verifier chooses between the direct candidate and the decomposed candidate. It is intentionally conservative: decomposition remains the default because it improves multi-hop retrieval, and the verifier switches to the direct answer only when the direct path is confident and the decomposition path shows signs of instability. The final implementation additionally applies a conservative type-aware fallback that corrects a small number of answer-type mismatches without rerunning generation or training a router.

Signals. The verifier uses four query-level signals:

- **Direct-answer entropy** σ^{dir}: low entropy suggests the direct answer may be reliable.
- **Decomposition post-entropy** σ^{dec}: high entropy after decomposition and synthesis suggests the decomposed answer is unstable.
- **Structural validity**: unresolved placeholders, empty sub-answers, or malformed sub-questions indicate decomposition failure.
- **Candidate agreement**: normalised string agreement between a^{dir} and a^{dec} is treated as mutual support.

Rule. The verifier accepts agreement immediately; otherwise it chooses direct when the direct answer is confident and decomposition is unstable, and chooses decomposition in the remaining cases. The deployed thresholds are `direct_entropy_tau=0.4` and `decomp_post_entropy_tau=0.8`. The type-aware fallback is a deterministic refinement: it only changes the final route when answer type, direct entropy, decomposition post-entropy, and entity-fragment signals consistently indicate that the direct answer should be preferred. No dataset name, gold answer, trained classifier, or evidence-label supervision is used.

Evidence support. We also evaluated evidence-aware variants that score whether the final answer or chain answers are lexically supported by retrieved passages. These variants

are not part of the final method: they slightly improve HotpotQA but consistently hurt 2Wiki and MuSiQue, showing that lexical support is not a reliable global routing signal for compositional questions.

3.9 Training Protocol

Only the decomposer requires training. The answerer, synthesizer, and verifier are training-free at the routing stage.

3.9.1 Backbone and LoRA Configuration

- **Backbone:** Llama-3-8B-Instruct.
- **Adapter type:** LoRA [33] on all linear modules.
- **Rank, Alpha, and Dropout:** $r = 16$, $\alpha = 32$, dropout $= 0.05$.
- **Precision:** bf16.
- **Context length:** 1024 tokens.

3.9.2 Training Data

Table 3.3: Composition of the 12K decomposer training corpus.

Source	#samples	Decomposition source
MuSiQue	4,000	Gold sub-questions
2WikiMultihopQA	4,000	Evidence-triple conversion
PopQA	4,000	1-step triples
Total	12,000	

The PopQA component teaches the decomposer that a one-step plan is valid. This improves single-hop robustness, but Chapter 4 shows that it also creates negative transfer

on some multi-hop datasets. The verifier is introduced precisely because no single hard decomposition policy is optimal across all query types.

3.9.3 Optimisation

LoRA is trained with LLaMA-Factory [34]: effective batch size 16, learning rate 1×10^{-4} with cosine decay and 10% warmup, 3 epochs, and approximately 38 minutes on one A6000 GPU.

3.10 Inference Modes

Two adapter modes were evaluated during development:

Mode B. LoRA is enabled for decomposition and synthesis but disabled for sub-question answering.

Mode C. LoRA is enabled for decomposition and sub-question answering. This is the final default because it outperformed Mode B in pilot experiments.

For final synthesis, the adapter is disabled (**synthOFF**) because the base instruction model produces shorter and more stable final spans.

3.11 Implementation Summary

Algorithm 1 summarises the final inference loop. Code lives in the `seek` module used by the main methodology code.

The next chapter evaluates this final dual-path system against DeepRAG under the honest wiki-21M retrieval protocol and reports the older distractor-pool experiments separately as development diagnostics.

Algorithm 1 SEEK v2 inference for one question q

Require: question q, corpus $\mathcal{P}$, entropy threshold τ, top-k

$(a^{\text{dir}}, \sigma^{\text{dir}}) \leftarrow \textsc{DirectAnswer}(q, r(q, \mathcal{P}, k))$
$\{s_1, \ldots, s_n\} \leftarrow \mathcal{D}_\theta(q)$
$A \leftarrow \emptyset$
for $i = 1$ to n **do**
    $s_i \leftarrow \textsc{Resolve}(s_i, A)$
    $(\tilde{a}_i, \sigma_i) \leftarrow \mathcal{A}_\theta(s_i, \emptyset)$
    if $\sigma_i \geq \tau$ **then**
        $c_i \leftarrow r(s_i, \mathcal{P}, k) \cup r(q, \mathcal{P}, k)$
        $a_i \leftarrow \mathcal{A}_\theta(s_i, c_i)$
    else
        $a_i \leftarrow \tilde{a}_i$
    end if
    $A[i] \leftarrow a_i$
end for
$(a^{\text{dec}}, \sigma^{\text{dec}}, z^{\text{dec}}) \leftarrow \textsc{Synthesize}(q, \{(s_i, a_i)\}_{i=1}^{n})$
$\hat{a} \leftarrow \textsc{Verifier}(a^{\text{dir}}, a^{\text{dec}}, \sigma^{\text{dir}}, \sigma^{\text{dec}}, z^{\text{dec}})$
return $\hat{a}$

Chapter 4 Experiments

This chapter evaluates the final SEEK system under the same honest wiki-21M retrieval protocol used by DeepRAG and compares it with representative reasoning and adaptive-retrieval baselines. After the main results, the chapter examines why the system works: whether decomposition actually improves retrieval, how much of the remaining MuSiQue gap comes from wrong intermediate answers rather than from decomposition wording, and whether retrieval or the entropy gate could be removed without losing accuracy. The later sections then look at the two answer paths in more detail. They measure how often each path is selected and where each path fails, compare SEEK's errors with DeepRAG's on the same questions, and ask how far entropy can be trusted as a confidence signal.

4.1 Experimental Setup

4.1.1 Datasets

We evaluate on HotpotQA, 2WikiMultihopQA, MuSiQue, and PopQA. HotpotQA, 2WikiMultihopQA, and MuSiQue test multi-hop reasoning, while PopQA tests single-hop factual recall and exposes over-decomposition on simple entity-centric questions. To align with DeepRAG [11], all headline results use the first 1,000 questions of each development set without shuffling.

4.1.2 Retrieval Corpus

The main comparison uses the honest open-domain corpus: the shared Elasticsearch `wiki` index over `psgs_w100`, containing approximately 21M Wikipedia passages. We use BM25 with top-$k = 8$ for HotpotQA, 2WikiMultihopQA, and MuSiQue, and top-$k = 3$ for PopQA. SEEK applies a path-aware title policy: the direct path retrieves with the original question over paragraph text only, while the decomposition path retrieves

short sub-question queries over title and paragraph text. This policy is dataset-agnostic because it depends only on query type.

The final system further uses *original-question evidence union*. For each decomposed sub-question, the evidence context is the union of passages retrieved by the sub-question and passages retrieved by the original question. This preserves the global context that can be lost when a long question is decomposed into short sub-questions.

4.1.3 Metrics and Baselines

We report SQuAD-style exact match (EM) and token-level F1 after standard normalisation. When a dataset provides aliases, the maximum score over aliases is used. The main table includes representative Llama-3-8B baselines reported by DeepRAG [11]: CoT and CoT-Retrieve as reasoning baselines, FLARE [6] and DRAGIN [7] as training-free adaptive retrieval baselines, and DeepRAG as the strongest learned adaptive baseline. Internal comparisons include the path-aware baseline, the decomposition-path original-question evidence union ablation, and the final SEEK system with type-aware verifier replay.

4.1.4 Implementation Details

All SEEK results use Llama-3-8B-Instruct with the 12K three-source LoRA decomposer trained on MuSiQue, 2WikiMultihopQA, and PopQA. The decomposition path uses Mode C, maximum token entropy threshold $\tau = 0.3$, and synthesis with the LoRA adapter disabled. The final verifier uses the original v2 confidence signals plus a conservative type-aware fallback. This fallback changes only the final choice between the direct and decomposed candidates; it does not rerun generation and does not train an additional model.

4.2 Main Results

Table 4.1 reports the final honest wiki-21M results. To keep the comparison aligned with the SEEK evaluation protocol, the table shows the four overlapping datasets used in

this thesis and omits CAG and WebQuestions, which were reported by DeepRAG but not evaluated by SEEK. The final SEEK system combines path-aware title retrieval, original-question evidence union, entropy-gated sub-question retrieval, and the type-aware training-free verifier.

Table 4.1: Main results on the first 1,000 questions of each dev set using the honest wiki-21M pool. Each cell reports EM/F1.

Type	Method	HotpotQA	2Wiki	MuSiQue	PopQA	Average
Reasoning	CoT	27.20/37.75	28.20/34.85	13.70/22.97	21.20/25.33	22.58/30.23
Reasoning	CoT-Retrieve	34.90/46.85	35.80/43.41	19.10/28.18	32.80/45.87	30.65/41.08
Adaptive	FLARE [6]	23.80/32.88	30.30/37.45	14.50/23.57	28.80/40.61	24.35/33.63
Adaptive	DRAGIN [7]	27.60/38.05	29.10/35.68	11.80/19.97	22.60/28.53	22.78/30.56
Learned adaptive	DeepRAG [11]	**40.70/51.54**	48.10/53.25	22.50/30.40	42.50/47.80	38.45/45.75
Ours	**SEEK final**	35.50/46.11	**50.00/56.58**	**24.70/33.06**	**51.90/54.71**	**40.52/47.61**

Note. External baseline numbers are the Llama-3-8B results reported by DeepRAG [11]. The average columns here are recomputed over the four datasets evaluated by SEEK.

The final SEEK system reaches 40.52 EM and 47.61 F1 on the four-dataset average, compared with DeepRAG's 38.45 EM and 45.75 F1, a gain of +2.07 EM and +1.86 F1 over the strongest external baseline. The margin over the reasoning baselines and the training-free adaptive baselines is considerably wider.

At the per-dataset level, SEEK wins on three of the four datasets. On 2Wiki it improves from DeepRAG's 48.10 EM to 50.00 EM, and on MuSiQue from 22.50 to 24.70 EM. The largest gain is on PopQA, where SEEK reaches 51.90 EM against DeepRAG's 42.50, which suggests the direct-vs-decompose verifier is especially useful when over-decomposition is risky.

HotpotQA is the exception. SEEK improves it from the path-aware baseline of 31.80 EM to 35.50 EM but still trails DeepRAG's 40.70 EM. HotpotQA is therefore the main remaining limitation and motivates the analysis in Sections 4.11–4.14.

4.3 Retrieval Recall

Before evaluating final answers, we first ask whether decomposition actually improves retrieval quality. Table 4.2 reports answer-containing hit@5: a query is counted as successful if at least one of the top five retrieved passages contains the gold answer or one of its aliases. Yes-or-no questions are excluded because the strings "yes" and "no" are not meaningful evidence strings.

Table 4.2: Answer-containing hit@5 under the final open-domain wiki retrieval condition. Original-query retrieval uses paragraph text only; decomposed sub-question retrieval uses title and paragraph text.

Dataset	N	Original	Decomposed	Union	Decomp − Original
2Wiki	917	31.73	62.38	**65.32**	+30.64
HotpotQA	941	46.55	56.54	**65.25**	+9.99
MuSiQue	1000	16.90	37.40	**41.60**	+20.50
PopQA	1000	68.10	68.70	**73.30**	+0.60

Decomposition substantially improves answer-containing retrieval on the multi-hop datasets: +30.64 points on 2Wiki, +20.50 on MuSiQue, and +9.99 on HotpotQA. PopQA is nearly unchanged because it is already single-hop. The union column is strongest on all four datasets, showing that decomposition should not discard the original question's global context.

4.4 Gold and Base Decomposition on MuSiQue

To separate the value of decomposition from the value of the trained decomposer, we run a focused MuSiQue retrieval-recall analysis using the first 1,000 development questions. This slice contains only 2-hop questions, which makes the official MuSiQue gold decomposition directly comparable with SEEK's predicted two-step chain. Table 4.3 reports answer-containing hit@5 under the same wiki-21M and path-aware retrieval con-

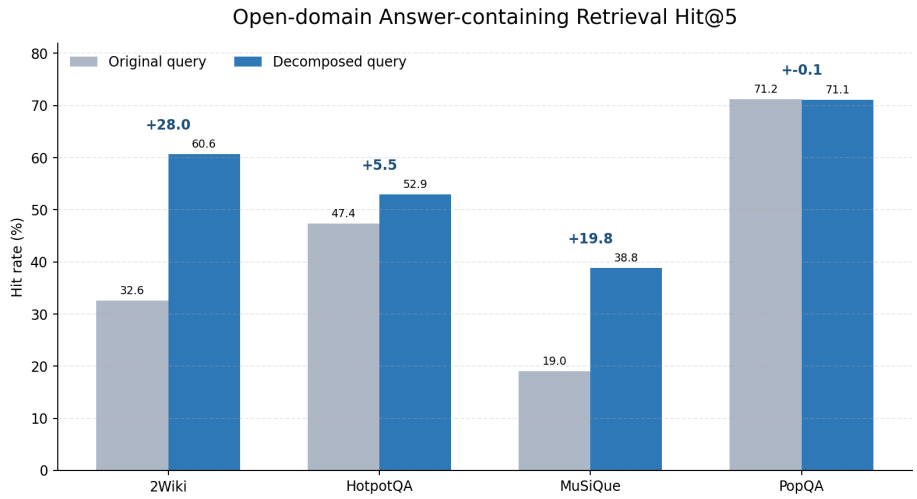


Fig. 4.1: Answer-containing retrieval recall under the open-domain wiki-21M condition. Decomposition improves multi-hop retrieval, while original-question evidence union further recovers context.

dition as Table 4.2.

The gap between original retrieval and SEEK retrieval is large: 19.0 to 38.8 hit@5. The oracle gold chain reaches 55.8, showing that decomposition has further retrieval headroom. However, the important diagnostic is that *Gold raw* collapses back to the original-query level. A 2-hop decomposition is useful only when the first-hop bridge entity is correctly resolved and substituted into the second query. SEEK's first-hop intermediate-answer EM is 41.9%. When the first hop is answered correctly, SEEK reaches 56.4 hit@5, essentially matching the gold chain on the same subset at 58.5. When the first hop is wrong, SEEK falls to 25.9 while gold stays at 53.8, so these questions are not intrinsically harder and their evidence remains reachable with the right bridge entity. The SEEK–gold gap is therefore almost entirely bridge-answer drift rather than poor decomposition wording.

The untuned base model also clarifies why the decomposer is trained. With the same prompt and no LoRA adapter, the base model produces a valid two-step decomposition for only 149 of 1,000 MuSiQue examples and has a 15.3% format-parsing failure rate. Its query text can retrieve well under oracle bridge substitution, but it is not structurally

Table 4.3: MuSiQue answer-containing hit@5 for original, SEEK, gold, and untuned-base decompositions.

Arm	Interpretation	Hit@5
Original question	No decomposition	19.0
SEEK	SEEK decomposition with predicted intermediate answer	38.8
Gold	Official MuSiQue decomposition with gold intermediate answer	**55.8**
Gold raw	Official decomposition without replacing `#1`	19.8
SEEK + gold bridge	SEEK sub-question text with gold intermediate answer	54.1
Untuned base + gold bridge	Llama-3-8B-Instruct without LoRA, using gold intermediate answer	52.9

reliable enough to deploy. LoRA training therefore buys step-count control, valid placeholders, and compact BM25-friendly query style, not merely better wording.

4.5 Retrieval Is Necessary After Decomposition

Decomposition alone does not produce factual knowledge. Table 4.4 compares three decomposition-path settings: never retrieve, entropy-gated retrieval at $\tau = 0.3$, and always retrieve.

Table 4.4: Ablation of retrieval after decomposition under honest wiki-21M. EM is reported for each answering strategy; retrieval rate is measured inside the decomposition path.

Dataset	Never retrieve	Gate $\tau = 0.3$	Always retrieve	Gain	Retrieval rate
HotpotQA	15.7	25.0	25.1	+9.3	92.7%
2Wiki	29.4	47.2	47.1	+17.8	95.2%
MuSiQue	9.6	24.7	25.0	+15.1	90.8%
PopQA	24.2	51.5	51.4	+27.3	99.3%

Retrieval after decomposition is necessary: the gate improves over never retrieve by

+9.3 to +27.3 EM. However, the gate and always-retrieve settings are almost identical in accuracy. This positions entropy gating as an efficiency mechanism rather than the main source of accuracy.

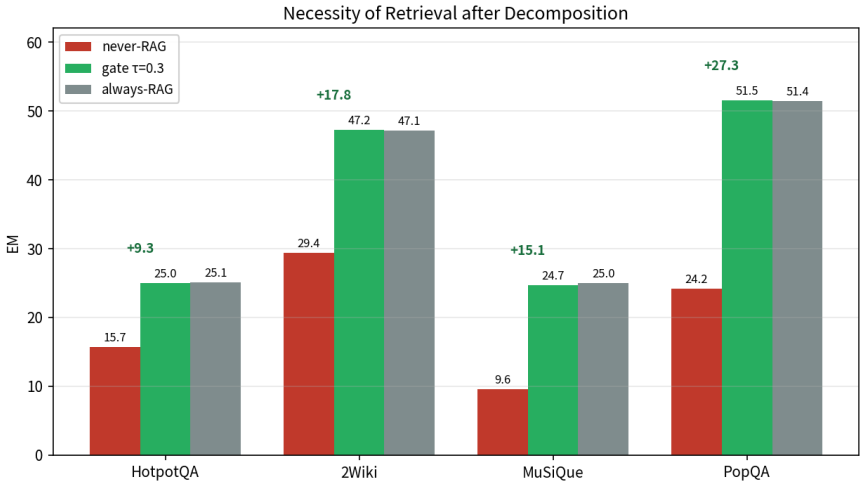


Fig. 4.2: Three-way decomposition-path ablation: never retrieve, entropy-gated retrieval, and always retrieve. The gate nearly matches always-retrieve accuracy while reducing some retrieval calls.

4.6 Direct-Only Confidence-Gated Baseline

The previous ablation shows that retrieval is necessary after decomposition. A separate question is whether decomposition itself is necessary, or whether the same entropy signal could simply gate retrieval on the original question. Table 4.5 compares a direct-only confidence-gated RAG baseline against the decomposition-path gate. The direct baseline first answers the original question without context, measures maximum token entropy, and retrieves the original question only when entropy exceeds a threshold. The table gives the matched threshold $\tau = 0.3$ and the best threshold selected retrospectively for the baseline.

Confidence-gated retrieval without decomposition plateaus at the single-query retrieval ceiling. Even when allowed to choose the best threshold after the fact, it remains far below decompose+gate on 2Wiki, MuSiQue, and PopQA. HotpotQA is the exception

Table 4.5: Direct-only confidence-gated RAG compared with decompose+gate. EM is reported under honest wiki-21M.

Dataset	Direct never	Direct gate $\tau = 0.3$	Direct best τ	Decompose+gate	Decompose − direct best
HotpotQA	18.2	**30.7**	30.7	25.0	-5.7
2Wiki	12.6	18.5	18.6	**47.2**	+28.6
MuSiQue	5.9	10.4	10.8	**24.7**	+13.9
PopQA	22.4	37.1	39.2	**51.5**	+12.3

because its original questions often already contain enough lexical signal for BM25 to retrieve useful evidence. The final SEEK system handles this case by keeping the direct path and routing late; the full verifier reaches 35.5 EM on HotpotQA, above the direct-gate baseline.

4.7 Threshold Sensitivity

Table 4.6 evaluates the final dual-path system under different entropy thresholds. Higher thresholds reduce retrieval, but the average score declines.

Table 4.6: Final SEEK performance across entropy thresholds. Each score cell reports average EM; F1, and retrieval rate is the average decompose-path retrieval rate.

τ	Average EM; F1	Retrieval rate
0.3	**40.52**; **47.61**	94.16%
0.5	40.45; 47.52	91.32%
0.8	39.98; 47.04	85.97%
1.0	39.50; 46.63	82.35%

The chosen threshold $\tau = 0.3$ is accuracy-first. A threshold of 0.8 saves about eight retrieval-rate points but costs 0.54 average EM. The curve supports a conservative interpretation: under honest wiki-21M, most decomposed steps still require retrieval, and tightening the gate eventually hurts multi-hop datasets such as 2Wiki and MuSiQue.

4.8 Path-Aware Title Retrieval

Original questions and decomposed sub-questions are different query types. A short query such as `Houston Baptist University founded` benefits from title matching because the title often identifies the relevant Wikipedia article. A long original question, however, can contain multiple entities and conditions; title matching can overemphasise a stray entity and retrieve the wrong article. Table 4.7 compares title-field policies under the same verifier setting.

Table 4.7: Title-field matching ablation on honest wiki-21M. Each cell reports EM; F1.

Dataset	Off	Global	Path-aware
PopQA	51.2; 53.5	51.5; 53.9	**51.8**; **54.2**
MuSiQue	22.5; 30.5	24.5; 33.1	**24.6**; **33.5**
2Wiki	45.2; 51.3	45.9; 52.5	**46.9**; **53.5**
HotpotQA	31.9; 41.2	30.5; 39.9	**31.8**; **41.2**

Path-aware title retrieval is the most stable option. It keeps the title benefit for short sub-questions while avoiding the HotpotQA drop caused by enabling title matching on long original questions.

4.9 Original-Question Evidence Union

The preceding retrieval results suggest that decomposition improves precision but may lose global context. Original-question evidence union addresses this by adding the original-question top-k evidence to each decomposed step. Since this operation only changes the decomposition path, the clean ablation compares the decomposed answer candidate with and without union, rather than mixing in questions for which the final verifier selected the direct path. Table 4.8 reports this path-local effect.

The path-local improvement is largest on HotpotQA and 2Wiki, precisely where

Table 4.8: Decomposition-path effect of adding original-question evidence to each decomposed sub-question context. Each cell reports EM/F1.

Dataset	Decomp w/o union	Decomp + union	Change
HotpotQA	25.00 / 33.42	**30.50 / 39.49**	+5.50 / +6.07
2Wiki	47.20 / 53.63	**50.20 / 56.65**	+3.00 / +3.03
MuSiQue	24.70 / 33.10	**25.50 / 33.40**	+0.80 / +0.30
PopQA	51.50 / 53.68	**52.90 / 55.20**	+1.40 / +1.52
Average	37.10 / 43.46	**39.78 / 46.19**	+2.68 / +2.73

short sub-questions can remove comparison targets or entity constraints from the original question. The full-system score also improves after union, but that number includes the verifier's direct-vs-decompose choices; it is therefore a final-system result, not the clean isolation of the union operation. In short, decomposition should produce sharper sub-queries without forcing the lower path to forget the original question.

4.10 Final Path Selection Rate

Original-question evidence union shares original-question passages with the decomposition path, but it does not collapse SEEK into a single path. The final verifier still selects either the direct answer candidate or the decomposed answer candidate. Table 4.9 reports the selected path rates in the final traces.

Table 4.9: Final verifier path selection rate after original-question evidence union and type-aware fallback.

Dataset	Direct path selected	Decomposition path selected
HotpotQA	274 / 1000 (27.4%)	726 / 1000 (72.6%)
2Wiki	65 / 1000 (6.5%)	935 / 1000 (93.5%)
MuSiQue	182 / 1000 (18.2%)	818 / 1000 (81.8%)
PopQA	37 / 1000 (3.7%)	963 / 1000 (96.3%)
Pooled	558 / 4000 (14.0%)	3442 / 4000 (86.1%)

These rates show that the system remains decomposition-dominant overall, while still preserving a direct path for cases where decomposition loses important context or produces an unstable answer. The shared original-question evidence is a context-preservation mechanism; the answer candidates and final routing decision remain path-specific.

4.11 Direct-vs-Decompose Correctness Distribution

Table 4.10 compares answer correctness across the direct and decomposition paths in the final traces. This analysis asks whether decomposition is always better, or whether the two paths solve different questions.

Table 4.10: Answer-level correctness distribution across direct and decomposition paths in the final traces.

Dataset	Both correct	Direct only	Decomp only	Both wrong	Decomp only − Direct only
HotpotQA	192	111	113	584	+2
2Wiki	141	39	361	459	+322
MuSiQue	71	31	184	714	+153
PopQA	350	28	179	443	+151

The distributions show that decomposition is useful but not universally dominant. On 2Wiki, MuSiQue, and PopQA, the decompose-only column is much larger than the direct-only column, confirming the value of decomposition. HotpotQA is different: direct-only and decompose-only are almost tied. This explains why HotpotQA remains difficult and why the final system must preserve both paths. The large both-wrong counts also show that many errors cannot be fixed by routing alone; they require better retrieval, reranking, decomposition quality, or final synthesis.

4.12 Decomposition-Path Failure Taxonomy

To understand what remains after applying decomposition, evidence union, and verifier routing, we classify decomposition-path errors on HotpotQA, 2Wiki, and MuSiQue.

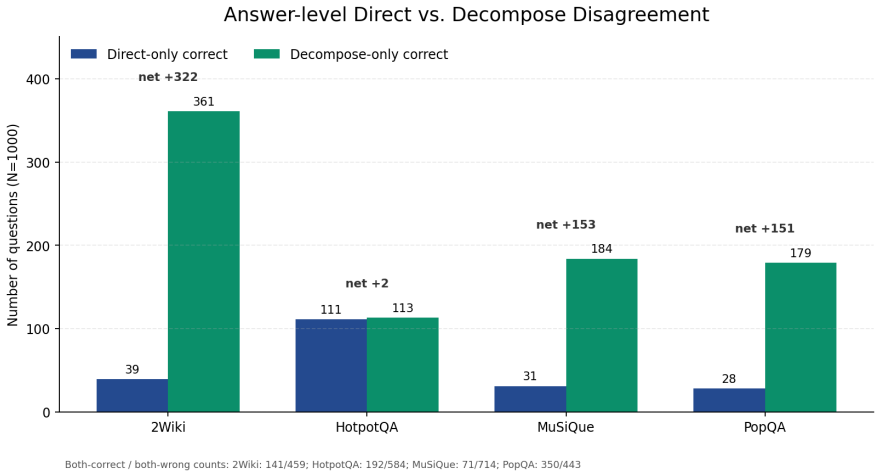


Fig. 4.3: Direct-vs-decompose correctness counts in the final traces. Decompose-only wins dominate on 2Wiki, MuSiQue, and PopQA, while HotpotQA is balanced.

Table 4.11 reports the primary error category among examples where the decomposed candidate is wrong.

Table 4.11: Failure taxonomy for decomposition-path errors. Values are shares within decomposed-candidate errors.

Category	HotpotQA	2Wiki	MuSiQue
Retrieval gap in both original and decomposed evidence	47%	64%	68%
Answer wrong despite retrieval coverage	28%	18%	12%
Synthesis failure	15%	17%	18%
Constraint loss from decomposition	10%	1%	2%

The dominant failure is not malformed decomposition; it is retrieval coverage. Neither the original query nor the decomposed evidence union reaches enough supporting information in 47% to 68% of decomposed-candidate errors. Constraint loss is more dataset-specific: it is visible on HotpotQA, where short KG-style sub-questions can drop the comparison target or disambiguating condition. This explains why original-question evidence union helps HotpotQA most in Table 4.8. Synthesis failure remains a stable 15% to 18% slice, motivating the answer-type and value-to-entity guards described in Chapter 3.

4.13 Case-Level Comparison with DeepRAG

The taxonomy above describes SEEK's own decomposition failures. A natural follow-up question is how these failures relate to DeepRAG's on the same inputs. We join the DeepRAG traces with the final SEEK traces by question text on the three multi-hop datasets and compare per-question EM. Table 4.12 reports the resulting counts.

Table 4.12: Per-question correctness of DeepRAG and SEEK on the same first 1,000 dev questions. The SEEK-only column is analysed in Table 4.13.

Dataset	Both correct	DeepRAG only	SEEK only	Both wrong
HotpotQA	257	139	98	506
2Wiki	340	162	160	338
MuSiQue	154	83	92	669

The first observation is that the two systems are complementary rather than ordered. The DeepRAG-only and SEEK-only columns are of similar size on every dataset, so the comparison does not show one method dominating the other per question. Its value lies in showing that the two systems fail in different ways.

To characterise those differences, we inspect the DeepRAG traces inside the SEEK-win set and count recurring failure signals. Table 4.13 reports the counts; a single case can match several signals.

Three patterns account for most of the substantive losses. The first is over-decomposition on 2Wiki comparison questions, where 39 of the 160 SEEK wins involve DeepRAG chains of five or more steps. A representative case asks which film has the earlier-born director, *Buffet Froid* or *Lost in Africa*. DeepRAG produces six steps but retrieves only once: it resolves the first director correctly, then answers the remaining steps from parametric memory, hallucinates the second director, and finally appends two redundant comparison steps that restate the original question. The wrong parametric comparison in those terminal steps flips an otherwise recoverable chain. SEEK's decomposer emits exactly the four

Table 4.13: Failure signals observed in DeepRAG traces on questions that SEEK answers correctly and DeepRAG does not. Counts are non-exclusive.

Signal	HotpotQA (n=98)	2Wiki (n=160)	MuSiQue (n=92)
Format near-miss, F1 $\geq$ 0.5	21	17	14
Degenerate final answer	24	29	27
Over-decomposition, $\geq$5 steps	5	39	5
Decomposition spiral	31	11	24
No retrieval in the whole chain	13	55	0

factual lookups, retrieves for each, and leaves the comparison to the synthesis stage.

The second pattern is a decomposition spiral. DeepRAG generates each follow-up question conditioned on the chain answered so far, so a wrong intermediate answer can rewrite the rest of the plan. In one HotpotQA case, the model resolves the stage name Aladin to the wrong person and then spends five further steps trying to make that person fit the question, with later follow-up fields visibly containing fragments of the earlier erroneous reasoning. This contamination appears in roughly a quarter to a third of the SEEK-win cases on HotpotQA and MuSiQue. SEEK's decomposition is produced up front from the original question alone, so a wrong bridge answer degrades one retrieval query but cannot corrupt the structure of the remaining sub-questions.

The third pattern is structural mis-decomposition. Asked who sings *Home Alone Tonight* with the singer of another song, DeepRAG splits the question into two independent singer lookups, drops the duet relation, and returns the same artist for both. SEEK's placeholder binding keeps the relation explicit in the second sub-question and recovers the duet partner. On 2Wiki, an additional dataset-specific weakness appears: in 55 of the 160 SEEK wins, DeepRAG's retrieve-or-answer policy chose the parametric option at every step and hallucinated the whole chain. SEEK's entropy gate keeps step-level retrieval above 90% on the multi-hop datasets (Table 4.4), so a fully parametric chain is rare.

These counts should be read with two caveats. About 15% to 21% of the SEEK-win set are format near-misses in which DeepRAG's answer is essentially correct but fails ex-

act match, such as *Murray Hill, New Jersey* against the gold *Murray Hill*. The comparison is also symmetric: within its own win set, SEEK's decomposition is sometimes poor as well, and the question is rescued by the direct path or by original-question evidence union rather than by the decomposition itself. On HotpotQA, 29 of the 98 SEEK wins come from the direct candidate. The DeepRAG-only column contains failures of the mirrored kind, which is consistent with the dual-path design rather than evidence against it.

4.14 Oracle Direct-vs-Decompose Headroom

Table 4.14 reports an oracle that chooses between the direct and decomposed answer using the gold answer. This oracle is not deployable, but it quantifies the remaining headroom in the routing decision.

Table 4.14: Oracle choice between direct and decompose in the final traces. Current router is the final SEEK verifier output.

Dataset	Direct EM	Decomp EM	Current router	Oracle	Oracle gap
HotpotQA	30.30	30.50	35.50	**41.60**	+6.10
2Wiki	18.00	50.20	50.00	**54.10**	+4.10
MuSiQue	10.20	25.50	24.70	**28.60**	+3.90
PopQA	37.80	52.90	51.90	**55.70**	+3.80

The oracle gap gives a precise interpretation of the remaining limitation. SEEK has not completely solved direct-vs-decompose routing; rather, it exposes the problem and recovers part of the available headroom with a training-free verifier. HotpotQA has the largest gap, +6.10 EM, because neither fixed path dominates.

4.15 Entropy as a Confidence Signal

Entropy remains useful, but it should not be overstated. Across pooled self-consistency probes, higher maximum entropy is associated with a higher error rate, with Spearman

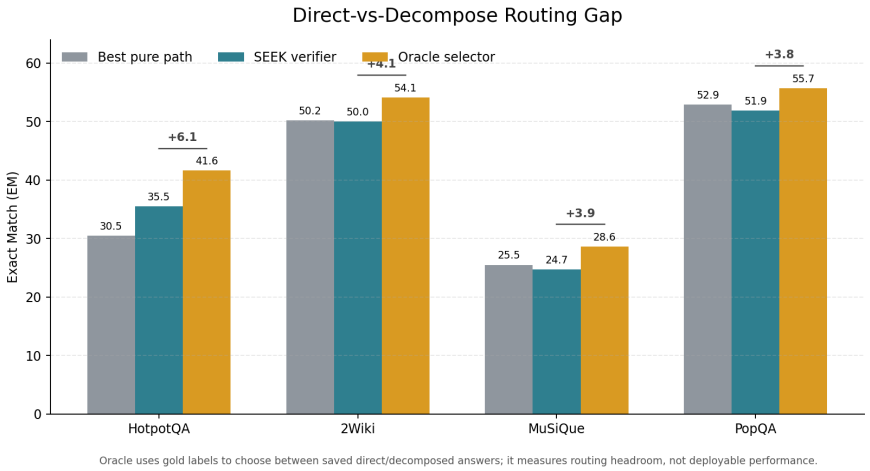


Fig. 4.4: Oracle direct-vs-decompose headroom in the final traces. The gap measures routing headroom only; both-wrong examples remain outside the verifier's reach.

correlation +0.326 and AUROC 0.686. In the gate-signal replay, maximum entropy also has the best macro AUROC among eight candidate retrieval-gate signals. These results support max entropy as a sub-question-level retrieval gate.

At the same time, entropy is not a correctness guarantee. Table 4.15 compares question-level direct-path entropy between examples where both paths are correct and examples where both paths are wrong. The separation is real: pooled AUROC is 0.782 and the mean direct entropy rises from 0.485 to 1.404. However, each dataset still contains confidently wrong examples whose entropy is at or below the median entropy of both-correct examples.

Table 4.15: Direct-path entropy separation between both-correct and both-wrong examples. Lower entropy means higher confidence.

Dataset	Mean entropy both-correct	Mean entropy both-wrong	AUROC	Confidently wrong rate
HotpotQA	0.407	1.151	0.750	18.3%
2Wiki	0.796	1.495	0.688	28.8%
MuSiQue	0.742	1.689	0.752	16.5%
PopQA	0.351	1.185	0.819	9.5%
Pooled	0.485	1.404	0.782	–

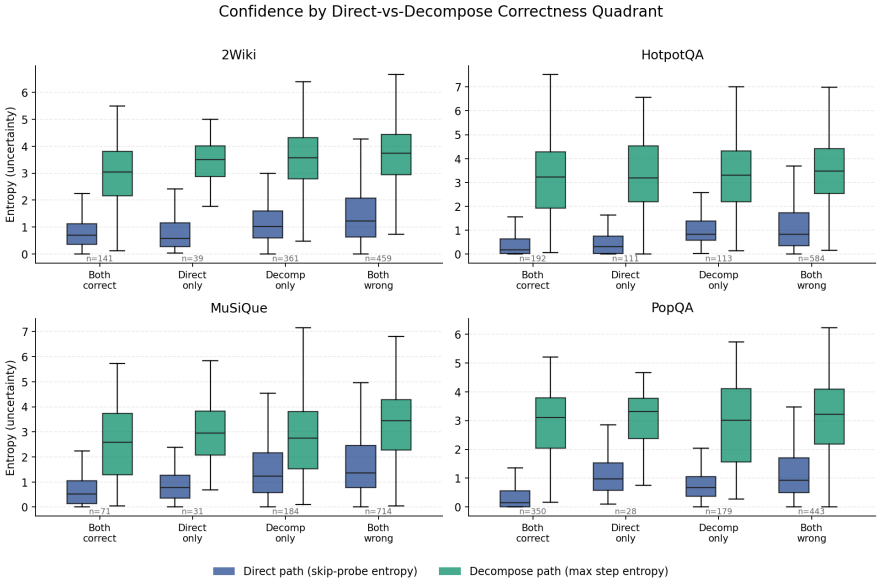


Fig. 4.5: Entropy by direct-vs-decompose correctness quadrant. Both-correct examples are more confident on average, but low entropy does not eliminate both-wrong cases.

The AUROC of 0.782 should be read as a population-level statistic rather than as the accuracy of a per-question detector. The medians are well separated, with pooled values of 0.27 for both-correct and 1.12 for both-wrong, but the tails overlap heavily: on every dataset, the tenth percentile of the both-wrong group lies below the median of the both-correct group. Drawing one correct and one wrong example at random, the wrong one has the higher entropy 78% of the time, and that is all the AUROC states.

Table 4.16 looks at the same signal from the calibration side, binning questions into direct-path entropy quintiles and reporting the direct-path accuracy within each bin.

Table 4.16: Direct-path answer accuracy within entropy quintiles, from the lowest-entropy fifth (Q1) to the highest (Q5). Accuracy decreases monotonically on all four datasets.

Dataset	Q1 (lowest)	Q2	Q3	Q4	Q5 (highest)
HotpotQA	60.5	34.5	29.5	19.0	8.0
2Wiki	30.5	25.5	14.5	14.5	5.0
MuSiQue	22.0	13.5	7.0	5.5	3.0
PopQA	81.5	43.0	27.5	26.0	11.0

Accuracy falls strictly with entropy on all four datasets, which rules out the concern that low-entropy correct answers are merely lucky guesses; if they were, the low-entropy bins would not be systematically more accurate. The absolute levels are more sobering. On MuSiQue the most confident fifth of direct answers is correct only 22% of the time, and on 2Wiki 30.5%, so on the multi-hop datasets even the most confident direct answers are far from trustworthy. Only single-hop PopQA reaches a usable 81.5% in its most confident bin.

The same picture emerges when entropy is used as an error detector. If the flagging threshold is set so that only 10% of correct answers are falsely flagged, entropy catches between 36% and 49% of the wrong answers depending on the dataset. More than half of the wrong answers carry entropy as low as typical correct answers, and no threshold can reach them.

These numbers all concern the direct-path answer entropy. The decomposition-path signal is considerably weaker at the question level: the maximum step entropy sits near three nats in every correctness quadrant and separates both-correct from both-wrong with an AUROC of only 0.588. It remains informative as a per-step retrieval gate, which is how SEEK uses it, but it says little about whether the final decomposed answer is right. There is also a dataset-level failure in the gate setting itself: on 2Wiki, step entropy predicts whether a no-retrieval sub-answer is wrong with an AUROC of only 0.540, close to random, because many 2Wiki parametric errors are produced with full confidence.

A final question is whether some other confidence statistic would do better. We ran a battery of 42 signals on the direct path in the deployed setting, covering temperature-scaled and Rényi entropies, log-probability statistics, top-k mass and margin scores, self-consistency agreement, P(True)-style self-evaluation, and z-scored combinations, evaluated as error detectors on two 100-question pilot sets with paired bootstrap over 1,000 resamples. Table 4.17 lists the strongest signals.

A few alternatives edge out the deployed maximum-entropy signal, and the temperature-2 variant is significantly better on the 2Wiki pilot, though the difference stays within the

Table 4.17: Strongest signals from a 42-signal battery on the direct path, evaluated as error detectors on two pilot sets (2Wiki n=95, HotpotQA n=100). AUROC of predicting a wrong direct answer.

Signal	2Wiki	HotpotQA	Macro
Temperature-2 max entropy	0.691	0.667	0.679
Minimum top-5 probability mass	0.629	0.707	0.668
Rényi-0.5 max entropy	0.657	0.678	0.668
P(True) with self-consistency	0.599	0.695	0.647
Deployed max entropy	0.539	0.655	0.597

bootstrap confidence interval on HotpotQA. None of the 42 signals, however, behaves like a dependable per-question error detector: the best macro AUROC is 0.679, and at the 10% false-flag budget no signal catches even 40% of the wrong answers on both pilot sets. Given the small pilot size, we read this as evidence that the limitation belongs to confidence signals as a family rather than to the particular choice of maximum entropy, and the final system keeps the simpler deployed signal.

This is why SEEK uses entropy as one confidence signal inside a dual-path verifier rather than as a single global router. Confidence helps estimate whether a path may be reliable; it does not reliably identify which path is correct when the two paths disagree.

4.16 Distractor-Pool Development Results

Table 4.18 records the older per-question distractor-pool results. These numbers are useful for diagnosis, but they should not be compared directly with DeepRAG because gold passages are usually guaranteed to appear in the candidate pool.

The gap between distractor-pool and wiki-21M results confirms that retrieval corpus difficulty is a major confound. The final thesis therefore uses wiki-21M as the headline protocol.

Table 4.18: Development results under per-question distractor pools. Each cell reports EM; F1.

Dataset	HotpotQA	2Wiki	MuSiQue	PopQA
SEEK development	39.6; 52.4	45.0; 53.7	48.0; 57.1	30.3; 40.2

4.17 Summary

Under the honest wiki-21M protocol, SEEK exceeds DeepRAG on the four-dataset average and wins on 2Wiki, MuSiQue, and PopQA, with HotpotQA remaining the only dataset below DeepRAG. Decomposition is the main source of this result. It raises answer-containing recall strongly on the multi-hop datasets, and the gold MuSiQue decomposition shows that the recall ceiling extends to 55.8 hit@5, with most of the remaining gap explained by wrong intermediate answers rather than by decomposition wording. Retrieval stays necessary after decomposition, since removing it costs 9.3 to 27.3 EM, and the confidence gate cannot take decomposition's place: a direct-only confidence-gated baseline stays 12.3 to 28.6 EM below decompose-with-gate on 2Wiki, MuSiQue, and PopQA even with a retrospectively chosen threshold.

The analyses also mark the limits of each component. Entropy behaves as an efficiency signal rather than a correctness guarantee: gated retrieval nearly matches always-retrieve accuracy while saving some retrieval calls, yet confidently wrong answers survive any single threshold, and a battery of 42 alternative confidence signals does not change this picture. The direct and decomposition paths are complementary, with oracle routing worth a further 3.8 to 6.1 EM. The case-level comparison with DeepRAG shows the same complementarity across systems: DeepRAG's chain-conditioned decomposition suffers from over-decomposition, error spirals, and zero-retrieval hallucination that SEEK's up-front decomposition and near-always sub-question retrieval avoid, while DeepRAG wins a comparable number of questions of its own. Both-wrong examples remain the largest error mass on every dataset and call for better retrieval and synthesis rather than better routing alone.

Chapter 5 Discussion

This chapter interprets the final results from Chapter 4. Taken together, they argue against reducing adaptive RAG to a single retrieve-or-skip decision. In multi-hop QA, the system must also decide when to trust decomposition, when to preserve the original question, and when the remaining error is outside the reach of routing.

5.1 Cost–Accuracy Trade-off versus DeepRAG

DeepRAG [11] and SEEK both operate at a finer granularity than standard retrieve-then-read RAG, but the final experiments require a careful cost claim. SEEK should not be described as uniformly cheaper at inference time. In the retrieval-count audit over the honest wiki-21M traces, SEEK retrieves more often than DeepRAG on HotpotQA and 2Wiki, and less often on MuSiQue.

Table 5.1: Inference retrieval count audit against DeepRAG on the overlapping multi-hop datasets.

Dataset	SEEK EM	SEEK retrievals/q	DeepRAG retrievals/q	Difference
HotpotQA	34.3	2.078	1.716	+21%
2Wiki	49.9	2.341	1.451	+61%
MuSiQue	24.8	1.840	2.273	-19%

The efficiency claim that survives this audit concerns offline adaptation and routing training, not guaranteed lower inference retrieval. SEEK trains only a LoRA decomposer; its retrieval gate, direct-vs-decompose verifier, and type-aware fallback are training-free. DeepRAG instead learns a retrieval policy through trajectory construction, SFT, and DPO-style preference optimisation.

The gradient-training FLOPs are of the same order, so the claim is not that SEEK is an order of magnitude cheaper in raw backpropagation. What SEEK saves is the DPO

Table 5.2: Offline adaptation cost comparison between SEEK and DeepRAG. DeepRAG training wall-clock was not reproduced locally; its FLOPs are an order estimate from the published training pipeline.

Component	SEEK	DeepRAG
Trained component	KG-triple decomposer only	Retrieval policy model
Fine-tuning type	LoRA SFT, less than 1% trainable parameters	Full-parameter SFT plus full-parameter DPO
Training stages	One SFT stage	Two stages: SFT and 8-epoch DPO
Measured wall-clock	37.9 minutes on one GPU	Not reproduced locally; expected multi-GPU hours
Gradient FLOPs	2.40×10^{17} measured	About 3.3×10^{17} estimated
Data construction	Direct supervised decompositions	Binary-tree-search trajectories; 72B follow-up model plus up to 8 7B vLLM servers in the official construction pipeline
Dataset adaptation	Reuse the same training-free routing rules	Rebuild trajectories and retrain the retrieval policy for new settings

stage, the tree-search trajectory construction pipeline, and any learned retrieval-policy retraining. This is the practical meaning of the "lightweight" claim in this thesis.

5.2 Why Direct and Decomposition Paths Must Coexist

The answer-level 2x2 analysis shows that decomposition is valuable but not always dominant. On 2Wiki, MuSiQue, and PopQA, the decompose-only column is much larger than the direct-only column, which confirms that decomposition solves many examples the direct path misses. HotpotQA is the exception: direct-only and decompose-only are nearly tied. This means HotpotQA is not simply an always-decompose dataset; the original question often carries comparison targets or disambiguating clues that are weakened by decomposition.

This is why SEEK keeps both paths. The direct path preserves the full question context, while the decomposition path serves as a retrieval strategy that exposes intermediate entities. Each path fails on different examples, and that is what makes the verifier

necessary.

The case-level comparison with DeepRAG in Section 4.13 extends this point across systems. DeepRAG generates each follow-up question conditioned on the chain answered so far. This lets it adapt the plan during reasoning, but it also means a wrong intermediate answer can rewrite the rest of the plan: in the traces, follow-up questions issued after a wrong hop contain fragments of the earlier erroneous reasoning, and the chain keeps elaborating the wrong entity instead of recovering. SEEK's decomposition is produced up front from the original question alone. A wrong bridge answer still degrades the second-hop retrieval query, as the MuSiQue analysis in Section 5.3 shows, but it cannot corrupt the structure of the remaining sub-questions, and the direct path remains available as a fallback.

The retrieval decision differs in the same way. DeepRAG learns a retrieve-or-answer policy for each step, and on 2Wiki this policy becomes a liability: in 55 of the 160 questions that SEEK wins, DeepRAG retrieved nothing at all and answered the entire chain from parametric memory. SEEK's entropy gate keeps step-level retrieval above 90% on the multi-hop datasets, so a fully parametric chain is rare. The price, documented in Table 5.1, is that SEEK retrieves more often overall; the benefit is that it seldom hallucinates a whole reasoning chain.

5.3 What Training the Decomposer Buys

The MuSiQue gold-decomposition analysis clarifies the role of decomposer training. SEEK's predicted decomposition raises answer-containing hit@5 from 19.0 to 38.8, while a gold decomposition with a gold intermediate answer reaches 55.8. When the SEEK sub-question text is given the gold bridge entity, it reaches 54.1, nearly the same as gold. This means the decomposer's wording is not the main bottleneck; the main bottleneck is whether the first-hop bridge entity is answered correctly before the second query is formed.

The untuned base-model comparison gives the other half of the story. The base Llama-3-8B-Instruct model can produce useful query text under oracle substitution, but it

produces a valid two-step decomposition for only 149 of 1,000 MuSiQue examples and has a 15.3% parsing failure rate. LoRA training therefore buys structural reliability: valid step counts, valid placeholders, and compact query style. That reliability is what makes the decomposition path executable at inference time.

5.4 Original-Question Evidence Union

Original-question evidence union is the most important methodological update in the final system. Decomposition improves answer-containing retrieval, but short sub-questions can lose the global context needed to interpret the relation correctly. By adding original-question evidence to each decomposed step, SEEK recovers part of this lost context without changing the decomposer or training a new retriever.

The cleanest comparison is within the decomposition path, because union changes only the decomposed sub-question context. Under this path-local ablation, HotpotQA's decomposed answer candidate improves from 25.00 to 30.50 EM, and 2Wiki improves from 47.20 to 50.20 EM. The full-system score also rises before the type-aware fallback, from 31.80 to 34.30 EM on HotpotQA and from 46.90 to 49.90 EM on 2Wiki, but that number additionally reflects the verifier's final path choices. This supports the thesis motivation directly: decomposition should make retrieval sharper, but it should not force the system to forget the original question.

5.5 Entropy: Useful Signal, Not Correctness Guarantee

Entropy remains useful for deciding whether a sub-question should retrieve. The never-retrieve, gate, and always-retrieve ablation shows that the entropy gate closely matches always-retrieve accuracy while saving a modest amount of retrieval. The gate-signal replay also shows that maximum entropy is the strongest of the tested uncertainty signals on average.

However, entropy is not a correctness guarantee. Both the self-consistency analysis and the direct-vs-decompose quadrant analysis show a stable confident-wrong region: in

the quadrant analysis, 9.5% to 28.8% of both-wrong examples have direct-answer entropy at or below the median entropy of both-correct examples.

A compact way to state these findings is that entropy measures the dispersion of the answer distribution, not the correctness of the reasoning behind it. The most common confident error in our traces is an incomplete reasoning path: the model stops at the first-hop bridge entity, or returns an attribute where the question asks for a comparison winner, and it does so with the same low entropy as a fully correct answer. In a separate diagnostic experiment that sampled each question fifteen times without retrieval, 1,538 of 4,000 questions produced fifteen identical answers and hence zero entropy, yet 40.3% of those answers were wrong. No threshold on an entropy-like statistic can flag such cases, because there is no dispersion to detect. The mirror group of low-confidence correct answers also deserves care: nearly half of them pass only through token-F1 overlap rather than exact match, so they are officially correct but not always semantically correct, and high entropy is an equally poor reason to discard an answer.

Entropy is also of little use for choosing between the two paths. When the direct and decomposed answers disagree, picking the path with lower entropy performs no better than chance once the two entropy scales are normalised, with per-dataset accuracy between 45% and 69%. The verifier's advantage on these disagreements comes from its structural rules and its decomposition prior, not from comparing confidence values.

This explains why the final system uses entropy inside a broader verifier rather than as a single hard router. Entropy helps locate uncertainty, but the direct-vs-decompose choice also needs answer type, decomposition stability, and candidate agreement. Even then, the remaining confidently wrong cases point toward future evidence-checking or self-consistency methods rather than another global threshold.

5.6 Remaining Headroom and Remaining Problems

The oracle analysis gives a measured upper bound for better routing. In the final traces, oracle direct-vs-decompose routing reaches 41.60 EM on HotpotQA, 54.10 on 2Wiki, 28.60 on MuSiQue, and 55.70 on PopQA. The current verifier still leaves 3.8 to

6.1 EM of routing headroom.

This headroom should be interpreted carefully. Oracle routing is not deployable because it uses the gold answer after the fact. It also cannot fix examples where both paths are wrong. The final traces contain 584 both-wrong HotpotQA examples, 459 2Wiki examples, 714 MuSiQue examples, and 443 PopQA examples. These examples require improvements below the router: better retrieval, reranking, bridge-answer propagation, or final synthesis.

The failure taxonomy makes the bottleneck concrete. Among decomposition-path errors, retrieval gaps account for 47% of HotpotQA errors, 64% of 2Wiki errors, and 68% of MuSiQue errors. Synthesis failures contribute another 15% to 18%. Constraint loss is smaller overall but important on HotpotQA, where it explains why original-question evidence union gives the largest gain. Thus the remaining problem after SEEK is not one problem; it splits into routing headroom, retrieval coverage, synthesis robustness, and confident wrong answers.

5.7 Threats to Validity

5.7.1 External Baseline Numbers Are Paper Numbers

The external baseline results in Table 4.1 are taken from the DeepRAG paper rather than reproduced in the exact same harness. Normalisation, prompting, and implementation details may differ slightly. To reduce protocol mismatch, SEEK uses the honest wiki-21M pool and the same first-1,000-question evaluation style, and the reported averages are recomputed only over the four overlapping datasets.

5.7.2 Dataset Slicing

All headline results use the first 1,000 dev questions to align with DeepRAG. MuSiQue dev is sorted by hop depth, so this slice understates full-dev difficulty. Full-dev MuSiQue remains harder, especially for 4-hop questions.

5.7.3 Single Seed and Greedy Decoding

All numbers use greedy decoding without seed averaging. Small differences of 1 to 2 EM should be interpreted cautiously; larger effects such as PopQA's gain over DeepRAG, the 2Wiki improvement, and the original-question evidence union gains are more robust.

5.7.4 Routing Cannot Fix Both-Wrong Cases

The verifier can only choose between two existing candidates. If both direct and decomposed answers are wrong, routing cannot recover the example. This is the main reason oracle routing is an upper bound on routing quality rather than an upper bound on the whole QA system.

Chapter 6 Conclusions and Future Work

6.1 Summary of Contributions

This thesis introduced **SEEK**, a lightweight adaptive RAG framework for open-domain question answering. The main contribution is to frame adaptive RAG as a *decomposition-routing problem*: the system must decide not only when to retrieve, but also when to trust a decomposed reasoning path and when to preserve the original question.

The principal contributions are:

1. **Dual-path adaptive RAG.** SEEK generates both a direct answer and a decomposed answer, allowing the final system to compare original-question reasoning with step-by-step decomposition.

2. **KG-triple decomposition for sub-question retrieval.** The decomposer emits compact `entity >> relation` fragments with `#N` dependencies, making multi-hop retrieval queries explicit and executable.

3. **Entropy-gated sub-question retrieval.** Maximum token entropy provides a transparent and training-free signal for deciding whether each decomposed step should retrieve external evidence.

4. **Original-question evidence preservation.** The final system unions original-question evidence into each decomposed step, reducing context loss caused by overly short sub-questions.

5. **Training-free verification.** SEEK selects between direct and decomposed candidates using confidence, answer type, decomposition stability, and candidate agreement, without classifier training, tree search, DPO, or preference data.

6. **Honest wiki-21M evaluation.** Under the DeepRAG-aligned open-domain pool, SEEK reaches 40.52 average EM and 47.61 average F1, outperforming DeepRAG

on average and winning on 2Wiki, MuSiQue, and PopQA.

6.2 Limitations

The principal limitations are:

- **HotpotQA remains below DeepRAG.** SEEK improves HotpotQA to 35.50 EM, but still trails DeepRAG's 40.70 EM. HotpotQA is the clearest case where direct and decomposed paths are complementary but the current verifier does not fully exploit the headroom.
- **Verifier headroom remains.** Oracle direct-vs-decompose routing still offers 3.8 to 6.1 EM over the current verifier, so routing is not solved.
- **Entropy is not a correctness guarantee.** Entropy is useful for sub-question retrieval gating, but 9.5% to 28.8% of both-wrong examples can look as confident as both-correct examples. A pilot battery of 42 alternative confidence signals did not find a decisive per-question error detector either (Section 4.15).
- **Bridge-answer drift limits decomposition.** On MuSiQue, SEEK nearly matches gold decomposition when the first-hop bridge entity is correct, but wrong intermediate answers drag the second query away from the target evidence.
- **Both-wrong cases require better retrieval and synthesis.** Routing cannot fix examples where both direct and decomposed candidates are wrong; retrieval gaps explain 47% to 68% of decomposition-path errors on the multi-hop datasets.
- **Single-seed greedy decoding.** Variance estimates are not reported and would strengthen small-difference claims.

6.3 Future Work

6.3.1 Stronger Verifier Signals

The largest immediate opportunity is improving the verifier. Candidate signals include self-consistency disagreement, top-1 versus top-2 logit margin, hidden-state variance, answer-type compatibility, and lightweight learned calibration from a small validation set. The signal battery in Chapter 4 suggests that gains from any single scalar signal are modest, so combining confidence with evidence-side checks is more promising than searching for another scalar. The goal is to approach the oracle direct-vs-decompose headroom without losing SEEK's interpretability.

6.3.2 Better Retrieval and Reranking

The large both-wrong counts show that routing alone is insufficient. Dense retrieval, hybrid BM25+dense retrieval, or reranking over the unioned evidence could improve cases where neither path retrieves enough supporting information.

6.3.3 Length-Conditioned Decomposer

A target-depth control token such as `[K=1]`, `[K=2]`, or `[K=4]` could reduce negative transfer between single-hop and multi-hop data. The verifier or a small complexity predictor could choose the target depth at inference time.

6.3.4 Multi-Sample Decomposition

Sampling multiple decompositions and selecting by final-answer confidence or agreement could reduce wrong-first-hop propagation. This is especially relevant because the MuSiQue analysis shows that bridge-answer drift, rather than decomposition wording alone, explains much of the gap between SEEK and gold decomposition. This would trade inference cost for robustness without requiring DPO.

6.3.5 Evidence-Aware Verification

Confidently wrong answers cannot be removed by another entropy threshold alone. Future verifiers should test whether the selected answer is externally supported by retrieved evidence, possibly with lightweight entailment checks, answer-type validation, or self-consistency over evidence snippets.

6.3.6 Span Calibration

A small post-training or post-processing module for terse answer spans could close EM-only failures while preserving F1. This is especially relevant for verbose entity descriptions and comparison questions.

6.4 Closing Remarks

This thesis argues that adaptive RAG should ask two questions: when should the system retrieve, and when should it trust decomposition? SEEK shows that a lightweight system can answer both questions with interpretable inference-time signals. Decomposition improves retrieval, entropy helps control sub-question evidence use, original-question evidence preserves global context, and a training-free verifier makes the final route explicit. The remaining oracle gap leaves a clear path for future work, but the central result is already useful: adaptive RAG can be competitive with heavier learned-policy systems without hiding its routing decisions inside another trained model.

References

[1] Y. Gao et al., "Retrieval-augmented generation for large language models: A survey", *arXiv preprint arXiv:2312.10997*, 2023.

[2] Y. Gao et al., "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions", *ACM Transactions on Information Systems*, 2025.

[3] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, "Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions", in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

[4] S. Jeong, J. Baek, S. Cho, S. J. Hwang, and J. C. Park, "Adaptive-rag: Learning to adapt retrieval-augmented large language models through question complexity", in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.

[5] Y. Wang, P. Li, M. Sun, and Y. Liu, "Self-knowledge guided retrieval augmentation for large language models", in *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.

[6] Z. Jiang et al., "Active retrieval augmented generation", in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[7] W. Su, Y. Tang, Q. Ai, Z. Wu, and Y. Liu, "Dragin: Dynamic retrieval augmented generation based on the real-time information needs of large language models", in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

[8] Z. Yao, W. Qi, L. Wang, L. Hou, and J. Li, "Seakr: Self-aware knowledge retrieval for adaptive retrieval augmented generation", in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[9] Y. Wang, Lu wei, and H. Ling, "Retrieval as a decision: Training-free adaptive gating for efficient rag", *arXiv preprint arXiv:2511.09803*, 2025.

[10] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-rag: Learning to retrieve, generate, and critique through self-reflection", in *International Conference on Learning Representations (ICLR)*, 2024.

[11] X. Guan et al., "Deeprag: Thinking to retrieve step by step for large language models", in *International Conference on Learning Representations (ICLR)*, 2026.

[12] Su et al., "Fast or better? balancing accuracy and cost in retrieval-augmented generation with flexible user control", *CoRR*, 2025.

[13] S. Min, V. Zhong, L. Zettlemoyer, and H. Hajishirzi, "Multi-hop reading comprehension through question decomposition and rescoring", in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019, pp. 6097–6109.

[14] T. Wolfson et al., "Break it down: A question understanding benchmark", *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 183–198, 2020.

[15] T. Yu, S. Zhang, and Y. Feng, "Auto-rag: Autonomous retrieval-augmented generation for large language models", *arXiv preprint arXiv:2411.19443*, 2024.

[16] X. Li et al., "Search-o1: Agentic search-enhanced large reasoning models", *arXiv preprint arXiv:2501.05366*, 2025.

[17] W. Feng, C. Hao, Y. Zhang, J. Song, and H. Wang, "Airrag: Activating intrinsic reasoning for retrieval-augmented generation via tree-based search", *arXiv preprint arXiv:2501.10053*, 2025.

[18] S. Kadavath et al., "Language models (mostly) know what they know", *arXiv preprint arXiv:2207.05221*, 2022.

[19] Z. Yin, Q. Sun, Q. Guo, J. Wu, X. Qiu, and X. Huang, "Do large language models know what they don't know?", in *Findings of the Association for Computational Linguistics: ACL 2023*, 2023.

[20] S. Kapoor et al., "Large language models must be taught to know what they don't know", *arXiv preprint arXiv:2406.08391*, 2024.

[21] X. Yin, X. Zhang, J. Ruan, and X. Wan, "Benchmarking knowledge boundary for large language models: A different perspective on model evaluation", *arXiv preprint arXiv:2402.11493*, 2024.

[22] Y. Yang, H. Yoo, and H. Lee, "Maqa: Evaluating uncertainty quantification in llms regarding data uncertainty", in *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025.

[23] L. Kuhn, Y. Gal, and S. Farquhar, "Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation", in *International Conference on Learning Representations (ICLR)*, 2023.

[24] S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal, "Detecting hallucinations in large language models using semantic entropy", *Nature*, vol. 630, pp. 625–630, 2024.

[25] R. Ren et al., "Investigating the factual knowledge boundary of large language models with retrieval augmentation", *arXiv preprint arXiv:2307.11019*, 2023.

[26] Z. Yang et al., "Hotpotqa: A dataset for diverse, explainable multi-hop question answering", in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2018.

[27] X. Ho, A.-K. Duong Nguyen, S. Sugawara, and A. Aizawa, "Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps", in *Proceedings of the 28th International Conference on Computational Linguistics (COLING)*, 2020.

[28] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, "Musique: Multihop questions via single-hop question composition", *Transactions of the Association for Computational Linguistics (TACL)*, 2022.

[29] A. Mallen, A. Asai, V. Zhong, R. Das, D. Khashabi, and H. Hajishirzi, "When not to trust language models: Investigating effectiveness of parametric and non-parametric memories", in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

[30] T. Kwiatkowski et al., "Natural questions: A benchmark for question answering research", *Transactions of the Association for Computational Linguistics (TACL)*, 2019.

[31] M. Joshi, E. Choi, D. S. Weld, and L. Zettlemoyer, "Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension", in *Proceedings of*

the 55th Annual Meeting of the Association for Computational Linguistics (ACL), 2017.

[32] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, "Squad: 100,000+ questions for machine comprehension of text", in *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2016.

[33] E. J. Hu et al., "Lora: Low-rank adaptation of large language models", in *International Conference on Learning Representations (ICLR)*, 2022.

[34] Y. Zheng et al., "Llamafactory: Unified efficient fine-tuning of 100+ language models", in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, 2024.

Appendices

Appendix A Supplementary Material

This appendix contains supplementary material such as full prompt templates, additional experimental results, dataset statistics, hyperparameter settings, and source code snippets.